Decomposing logophoric pronouns: a presuppositional account of logophoric dependencies*

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Abstract

Logophoric pronouns in West-African languages occur in attitude environments and are anaphorically linked to an attitude holder in a superordinate clause. This has motivated theorists to treat logophoric pronouns semantically as an obligatorily bound variable (bound from the edge of an embedded complement clause). Culy (1994) and Bimpeh and Sode (2021), however, pointed out that logophoric pronouns in Ewe do not behave like obligatorily bound variables, allowing both sloppy (bound) as well as strict (non-bound) readings in focus contexts involving 'only' and ellipsis. We strengthen this line of criticism by providing novel cross-linguistic data which indicates that logophoric pronouns in Ewe, Igbo and Yoruba support strict readings. We offer an alternative formal account to existing approaches which builds on Bimpeh et al. (2024) and can capture both strict and sloppy interpretations, while preserving the requirement that a logophoric pronoun be anaphoric to an attitude holder. The main novelty involves decomposition of logophoric pronouns into two syntactic components at LF—a variable which can in principle be free and refer strictly, and a semantic presuppositional feature LOG which can be ignored in ellipsis and focus sites, following similar ideas in the literature on pronominal features (Sauerland 2013). Our analysis implies that in terms of their syntactic and semantic make up, logophors are essentially no different from other pronouns, consisting of a referential index plus semantic features.

1 Introduction

Logophoric pronouns in some West-African languages are special anaphoric elements which typically occur in attitude contexts, and must refer back to the attitude holder (Clements 1975). In Ewe, for example, the logophoric pronoun (henceforth LOGP) *yè* normally appears in attitude ascriptions like in (1). For convenience the relationship between LOGP and its antecedent is represented with indexation.[1]

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[1]Glossing abbreviations follow the Leipzig glossing rules with the addition of LOGP = logophoric pronoun, ORDP = ordinary pronoun, RP = resumptive pronoun, PREP = preposition, RED = reduplication.

(1) Kòfí$_1$ bé **yè**$_1$ dzó. **Ewe**
Kofi say LOGP left
'Kofi said that he left.' (Clements 1975)

Ewe's LOGP is restricted to this kind of environment; in particular, it cannot be used in simple unembedded sentences to refer to an antecedent introduced in the previous sentence (or made contextually-salient otherwise). For instance, if Afi is the topic of conversation and we mention her having been at the party, then (2a) cannot be used as a follow-up; in such cases the ordinary 3rd person pronoun (henceforth ORDP) *é* must be used (2b).

(2) a. ***yè**$_1$ dzó **Ewe**
LOGP leave.
'She left.'

b. **é**$_1$ dzó
ORDP leave.
'She left.' (Pearson 2015: 78)

Clements (1975), one of the first to systematically study the phenomenon, thus characterizes Ewe's LOGP *yè* as an item "used exclusively to designate the person whose speech, thoughts, feelings or general state of consciousness are reported." In the literature since the 70's, the term 'logophoricity' has been applied to describe elements with a similar function of encoding sensitivity to the attitude of some perspective-holder (see Hagège 1974, Charnavel 2020a for French; Koopman and Sportiche 1989 for Abe; Nikitina 2012 for Wan; Kaiser 2018 for Finnish; Park 2018 for Korean; Sundaresan 2018 for Tamil; Kiemtoré 2022 for Jula; Newkirk 2019 for Ibibio; Schlenker 1999 for Amharic; and Sells 1987, Culy 1994, Stirling 1994, Güldemann 2003, Deal 2020 for cross-linguistic studies).

How is the dependency between LOGP and its antecedent encoded in the grammar? This paper is an attempt to advance towards an answer to this question, from the angle of the well-known strict-sloppy ambiguity of pronominal reference: our discussion and analysis will gear towards explaining why LOGPs allow strict readings in ellipsis and focus contexts (cf. Culy 1994, Bimpeh and Sode 2021).

As an overview, we provide new data from three languages—Ewe, Yoruba and Igbo—that confirm that LOGPs in these languages are ambiguous between sloppy and strict interpretations (our work also provides the first cross-linguistic study on *de se* readings of LOGPs which includes mistaken identity scenarios across several attitude predicates). We will discuss why the existence of the strict reading is problematic for current approaches (von Stechow 2003, Pearson 2015, a.o.), and will account for the problematic generalizations with a novel theory of the syntax-semantics interface of LOGPs. The main novelty is a decomposition of logophoric pronouns into two syntactic components at Logical Form (LF)—a variable and a semantic feature LOG which induces a presupposition, like other semantic features.

The paper is organized as follows. Section 2 provides general background on the basic distribution and interpretation of logophors in Ewe, Yoruba and Igbo. Section 3 presents our findings on strict and sloppy interpretations, and shows why they require modification of the existing accounts. Section 4 presents our new proposal for a syntax-semantics of LOGPs that can capture strict readings, including hitherto undescribed flavours of strict readings. Section 5 expands the proposal and accounts for long-distance LOGP dependencies as well as for the relationship between LOGP and the 1st-person pronoun in the languages of interest.

2 Logophors and their distribution and interpretation

We focus on three West African languages which have been observed to display logophoric pronouns: the Kwa language Ewe, and two Benue-Congo languages Yoruba and Igbo. Detailed language profiles can be found in the Appendix A.1. In section 2.1, we present the main distributional pattern of LOGPs in the languages under investigation, while section 2.2 is devoted to a discussion of the so-called *de se*-*de re* distinction in connection to logophors.

Unless indicated otherwise, all data in this paper come from original fieldwork. We elicited data from three Ewe speakers (two Anlo and one Ewedome), two Yoruba speakers and four Igbo speakers. All data was elicited via multiple Zoom sessions with each speaker, transcribed live by the experimenters and double-checked by the speakers. The elicitation language was English. Speakers' spontaneous comments on the reasoning behind their responses were also noted.[2]

2.1 Obligatory co-reference with the attitude holder

In (3), we present data for Ewe's LOGP *yè* embedded under the attitude predicates 'think', 'say', 'want', and 'hope'.[3] As is indicated by the indexation, *yè* must co-refer with the attitude holder in the matrix clause; it cannot refer to some other contextually salient individual.

(3) *Logophors in **Ewe***

a. Kòfí$_1$ súsú bé **yè**$_{1/*2}$ á dè Àfí.
Kofi think that LOGP IRR marry Afi
'Kofi thinks that he will marry Afi.'

b. Kòfí$_1$ gblɔ̀ bé **yè**$_{1/*2}$ á dè Àfí.
Kofi say that LOGP IRR marry Afi
'Kofi says that he will marry Afi.'

c. Kòfí$_1$ dʒí bé **yè**$_{1/*2}$ á dè Àfí.
Kofi want that LOGP IRR marry Afi
'Kofi wants to marry Afi.'

d. Kòkú$_1$ lè mɔ́-kpɔ́-m bé **yè**$_{1/*2}$ á dè Àfí.
Koku COP path-see-PROG that LOGP IRR marry Afi
'Koku hopes that he will marry Afi.'

[2]Given that we tested the distribution of LOGP and ORDP across several predicates, the actual number of test items per data point was never more than 2. As for *de se* / *de re* readings (section 2.2), strict/sloppy identity (section 3) and multiple embeddings (section 5.1), we tested the distribution of logophors and pronouns embedded under the verbs *say, think* and *hope* consistently with all our consultants. The verbs for *want* were tested with all speakers for de se readings. We also tested with the verbs for *promise* in all three languages (see Appendix A.2). For strict unknown identity, we only elicited judgements with the verbs for *think*.

[3]Although tense in Ewe (as well as in most Kwa languages) is not overtly marked, Ewe displays an irrealis marker *a* which expresses the possiblility of an event occuring in the future (Essegbey 2008). The irrealis marker *a* can be optionally added in (3a) and (3b) to express such a meaning. In (3c) and (3d), however, the marker occurs obligatorily. The environments where the irrealis marker occurs obligatorily roughly match the ones known in English as complements of control verbs; see also the discussion in Appendix A.3 and Grano and Lotven (2019) for Gengbe, an Ewe dialect spoken in Togo.

Yoruba and Igbo have LOGPs that exhibit comparable distributional properties to those of Ewe (Manfredi 1987, Hyman and Comrie 1981, Adésolá 2005, Lawal 2006). Examples (4) and (5) illustrate.[4]

(4) *Logophors in **Yoruba***

a. Adé$_1$ rò wípé **òun**$_{1/*2}$ fẹ́ Ọlá.
Ade think that LOGP marry Ola
'Ade thinks that he married Ola.'

b. Adé$_1$ sọ wípé **òun**$_{1/*2}$ fẹ́ Ọlá.
Ade say that LOGP marry Ola
'Ade said that he married Ola.'

c. Adé$_1$ fẹ́ wípé **òun**$_{1/*2}$ fẹ́ Ọlá.
Ade want that LOGP marry Ola
'Ade wants to marry Ola.'

d. Adé$_1$ ń rètí wípé **òun**$_{1/*2}$ máa fẹ́ Ọlá.
Ade PROG hope that LOGP FUT marry Ola
'Ade hopes that he will marry Ola.'

(5) *Logophors in **Igbo***

a. Ézè$_1$ chèrè nà **yá**$_{1/*2}$ lụ́rụ́ Àdá.
Eze think that LOGP marry Ada
'Eze thought that he married Ada.'

b. Ézè$_1$ sịrị̀ nà **yá**$_{1/*2}$ lụ́rụ́ Àdá.
Eze say that LOGP marry Ada
'Eze said that he married Ada.'

c. Ézè$_1$ chọ̀rọ̀ nà **yá**$_{1/*2}$ gà à-lụ́ Àdá.
Eze want that LOGP FUT PTCP-marry Ada
'Eze wants to marry Ada.'

d. Ézè$_1$ nwè-rè òlìlèányá nà **yá**$_{1/*2}$ gà à-lụ́ Àdá.
Eze be-RED hope that LOGP FUT PTCP-marry Ada
'Eze is hopeful that he will marry Ada.'

All three languages show identical co-reference patterns. The results of our elicitation sessions confirm previous reports in the literature. In the next section, we will report on another trademark property of logophors: obligatory *de se* readings.

[4] In contrast to Ewe, the forms for LOGP in Yoruba and Igbo are taken from the paradigm of strong pronouns. Hence, the forms themselves also occur in unembedded environments. The strong form *òun* in Yoruba is used in emphatic environments, while the strong form *yá* in Igbo is used additionally in object position (Pulleybank 1986, Manfredi 1987, Amaechi 2020).

2.2 The *de se*-*de re* distinction

For the elicitation of the data in this section, we used a binary acceptability judgment task designed with joint presentation for two target sentences (one with LOGP and one ORDP) to be judged against *de re* contexts: speakers were asked to express their acceptability judgments on both target sentences, but they were free to accept as felicitous both sentences, one sentence or none. Additionally, consultants were asked to judge the same target sentences against equivalent *de se* scenarios (given in Appendix A.2) to confirm our methodology and the data in section 2.1.

2.2.1 *de se*-only interpretation of logophors

It falls beyond the scope of this paper to provide a complete overview of the literature on LOGPs.[5] But one property of LOGPs which will be integrated into our analysis deserves elaboraton, and that is the *de se* reading of LOGPs. The so-called *de se*-*de re* distinction of pronominal reference in attitude contexts has to do with whether the attitude holder recognizes themselves as the real referent of the pronoun (Lewis 1979, Chierchia 1989, Kaplan 1989, Schlenker 1999, Pearson 2015, Park 2018, Patel-Grosz 2020, a.o.). Recently, Bimpeh et al. (2024) found evidence that LOGPs in Ewe, Yoruba and Igbo must meet this 'self-awareness' requirement. Specifically, Bimpeh et al. (2024) showed that LOGPs (but not ORDPs) are infelicitous in 'mistaken identity' scenarios as in (6), in which Donald Duck is referring to someone who, unbeknownst to Donald, is actually him.

(6) Mistaken Identity Context: (Bimpeh et al. 2024)
Donald Duck (DD) went to the grocery store to buy flour. He mistakenly put sugar in his cart. DD then saw a trail of sugar going up and down the aisles and thought that someone's bag had a hole in it and looked around for the guy. DD says: "I wonder who is losing sugar; certainly, the guy who is losing sugar is stupid, as he does not check his bag". Later he says: "Is it me the stupid guy who is losing sugar?" "No, because I did not buy sugar but flour".

a. Donald Duck súsú bé **#yè** / **é** dzɔ̀mɔ̀ví. **Ewe**
Donald Duck think that LOGP / ORDP stupid.person
'Donald Duck thinks that he is stupid.'

b. Donald Duck chèrè nà **#yá** / **ọ́** bụ̀ ónyéńzúzù. **Igbo**
Donald Duck think that LOGP / ORDP COP stupid.person
'Donald Duck thinks that he is stupid.'

c. Donald Duck rò pé **#òun** / **ó** jẹ́ òmùgọ̀. **Yoruba**
Donald Duck think that LOGP / ORDP COP stupid.person
'Donald Duck thinks that he is stupid.'

As with 'think' (6), logophors embedded under 'say' and 'hope' can only be read *de se*, highly suggested by the infelicity in a mistaken identity scenario in (7) and (8).

[5] Nor do we intend to cover in any depth the connection between LOGPs in West-African and other phenomena sometimes discussed under the general rubric of 'logophoricity': perspectival anaphora (Sundaresan 2018), exempt anaphora (Charnavel 2020a), and perhaps Indexical Shift (Deal 2020). But see Appendix A.4 for a comparison to the phenomenon of exempt anaphors.

(7) <u>*Mistaken Identity Context:*</u> (Bimpeh et al. 2024)
Elmo goes to visit Big Bird. While there, Big Bird shows him old paintings he found from back when Elmo was living there with him. After looking at several pictures, Elmo does not recognize one of the paintings which is particularly pretty. Elmo says: "I wonder who painted this. Certainly, the person who painted is a good painter". Later he says: "Is it me the good painter who painted this? No, because I am not very talented in painting".

a. Elmo bé **#yè** / **é** nyé nùtálá nyùìé a dé. **Ewe**
Elmo say LOGP / ORDP be painter good INDF
'Elmo said that he is a good painter.'

b. Elmo sị nà **#yá** / **ọ́** nà-ésè íhé ǹké ọ́má. **Igbo**
Elmo say that LOGP / ORDP IPFV-paint thing NKE good
'Elmo said that he is a good painter.'

c. Elmo so pé **#òún** / **ó** jẹ́ akunlé tí ó dára. **Yoruba**
Elmo say that LOGP / ORDP COP painter REL RP good
'Elmo said that he is a good painter.'

(8) <u>*Mistaken Identity Context:*</u>
Goofy is so drunk that he has forgotten he is a candidate in the election. He watches someone on TV and finds that that person is a terrific candidate, who should definitely be elected. Unbeknownst to Goofy, the candidate he is watching on TV is Goofy himself.

a. Goofy lè mɔ́-kpɔ́-m bé **#yè** / **é** à ɖù àkɔ̀-dádá lá dzí.
Goofy COP path-see-PROG that LOGP / ORDP IRR win vote-cast DEF post
'Goofy hopes that he will win the election.' **Ewe**

b. Goofy nà èlé ánya nà **#yá** / **ọ́** gà è-mérí n' àtúm̀vóòtù áhù.
Goofy have look eye that LOGP / ORDP FUT PTCP-win in election DEM
'Goofy hopes that he will win the election.' **Igbo**

c. Goofy ń rètí pé **#òun** / **ó** yoo borí nínú ìbò náà.
Goofy PROG hope that LOGP / ORDP FUT win in election DET
'Goofy hopes that he will win the election.' **Yoruba**

We take it then that LOGPs in these languages have a requirement for *de se* readings: LOGP can only refer to the attitude holder's 'recognized self'.

2.2.2 A note on methodology and conflicting generalizations

Our claim that LOGPs only have *de se* readings is controversial. While it converges with the results in Bimpeh (2019) for Ewe as well as with Adésolá (2005) and Anand (2006) for Yoruba (we are not aware of a previous discussion of *de se*-*de re* with respect to Igbo), Pearson (2015) by contrast reports that LOGP in Ewe also supports a *de re* reading, as most of the speakers in her study judged LOGP as felicitous and true in mistaken identity scenarios (see also O'Neill 2015, Satık 2021). These findings were partly confirmed by a quantitative questionnaire study on *de se*-*de re* readings by Bimpeh (2023) with 20 Ewe speakers. While 62.5% responses indicated that LOGP is infelicitous in mistaken identity contexts, 37.5% responses revealed that LOGP was accepted. This indeed suggests that LOGP at least sometimes seems to receive a *de re* reading. The questionnaire study in Bimpeh (2023), however, revealed several surprising results, most importantly ORDP was rejected by participants over half of the times (ORDP was judged infelicitous in mistaken identity scenarios 65.7% of the time). This outcome is unexpected under every account mentioned so far. Bimpeh (2023: 127) concludes that "a possible explanation [for the discrepancy in results between Bimpeh (2023) and Bimpeh (2019)–our addition] is that participants were mostly confused since they are not used to such scenarios", which is possibly why LOGP and the baseline ORDP version were both rejected over 50% of the time. The difference between one-on-one elicitation (the current study and Bimpeh 2019) and questionnaires (Bimpeh 2023) is that with the former, consultants can more comfortably be familiarized with the rather unusual mistaken identity contexts, and ask for clarification if necessary. Consequently, the one-on-one elicitation method used in Bimpeh (2019) achieved consistent results indicating exclusively-*de se* interpretation of LOGPs, which align with our findings.

It is important to try to clarify the nature of the disagreement, and to this end we now comment about the methodology we used to obtain our reults in (6)-(8) and explain how it differs from studies that arrived at conflicting conclusions to ours. Our technique to elicit *de re* readings is different from previous work on two points. First, as is clear from (6)-(8), and following Bimpeh (2023)'s idea, we let our speakers judge not just sentences with LOGP but also a minimally different variant with ORDP. This gave the speakers the opportunity to create a baseline, and it also allowed us to verify that speakers understood the context (they were asked to express their acceptability judgments on both versions, and were free to accept both sentences, one sentence or none). We think this is highly relevant, as mistaken identity contexts are rather difficult to take in (especially for non-linguists/semanticists). With the exception of Bimpeh (2023), previous work (Pearson 2015, O'Neill 2015, Satık 2020, 2021) did not test the ORDP version in mistaken-identity scenarios to establish a baseline. The fact that our speakers all accepted the ORDP version but rejected the LOGP version suggests that only the former is compatible with a *de re* reference.

Second, most of our mistaken-identity scenarios made sure that the target sentences are false on a *de se* reading. In the mistaken identity context in (6), for example, it is explicitely mentioned that Donald Duck does not self-ascribe stupidity. ("Is it me the stupid guy who is losing sugar? No because I did not buy sugar but flour.") No previous work we are aware of (Pearson 2015, O'Neill 2015, Satık 2020, 2021, Bimpeh 2023) made it unambiguously clear in the description of the context that a *de se* interpretation is false. We believe it is important do to so, in order to make sure that upon judging the LOGP sentence, speakers do not apply some charity principle and mentally modify the mistaken-identity context ever so slightly so as to make it possible for the sentence to be true on a *de se* reading. Whenever our speakers were asked if the sentence is true or false against a mistaken-identity context that made it sufficiently clear that a *de se* interpretation is false, our speakers indicated to us that the sentence is false.[6] For more discussion on the methodology, see Appendix A.2.

[6]Given that Pearson 2015 and other works reached the opposite conclusion to ours for Ewe (using different methodology), a reviewer suggests that one cannot rule out the possibility that there is a dialect split or cross-speaker variation in Ewe. On this point we side with Pearson (2022) who contends that it is unclear what sort of evidence in the input could trigger the acquisition of different grammars by different speakers on the *de re* property of LOGPs. Moreover, we hope that our methods for eliminating possible confounds will prove useful in clarifying the empirical picture (for Ewe). But if it turns out that a genuine variation is what underlies the disagreement in the literature, then our analysis from hereon should be taken to apply only to those speakers (like ours) who only accept *de se* readings of LOGPs; if there are speakers whose grammar truly allows *de re* construals of LOGP, our account would have to be enriched with further mechanisms to accommodate them.

3 Strict readings of logophors

3.1 The Simplex Binding account of logophors and a problematic prediction

In the formal semantic literature on LOGPs, it has become standard to capture their basic distributional facts—namely, the requirement for (*de se*)-coreference with the attitude holder—by assuming that LOGPs are simplex bound variables, bound from the left periphery of complement clauses. This is the view taken for example by Schlenker (2003), von Stechow (2004), Heim (2005), Pearson (2015). We will call this the SIMPLEX BINDING approach to LOGPs. Let us briefly go over how this approach works, as it will later be criticized based on the availability of strict readings of LOGPs. We use the account in Pearson (2015) to illustrate the approach.[7]

Pearson (2015), following von Stechow (2003), assumes that LOGP is like a standard pronoun in being interpreted as a bare variable (via an assignment function as usual), but it comes with a syntactic feature LOG whose purpose is to make sure that that variable ends up being bound by a λ-operator at the edge of an embedded clause (technically by feature 'checking' between [LOG] and the matching λ). To illustrate, the LF representation of *Kofi says that* LOGP *will marry Afi* is in (9a), where [LOG] enforces index matching between the variable and the binder at the edge of the CP (the boldfaced λ operator). This syntax is coupled with a semantics that assigns the embedded clause a property meaning (type $\langle \texttt{e,st} \rangle$), and an appropriate meaning for attitude predicates like *say* which involves quantification over *centered worlds* (Lewis 1979), see (9b).

(9) *Simplex Binding account of* LOGP *(based on Pearson 2015)*

a. Syntax:
Kofi says that $\big[\boldsymbol{\lambda x_1}\, \lambda w\ \ [\underbrace{\boldsymbol{x_{1_{[\text{LOG}]}}/^{*}x_{2_{[\text{LOG}]}}}}_{\text{LogP}}$ will marry Afi$]\big]$

b. Semantics:
$[\![\text{say}]\!]^w = \lambda P \lambda x.\ \forall \langle w', x' \rangle \in \text{SAY}_{x,w},\ P(x')(w') = 1.$
$\text{SAY}_{x,w} := \{\langle w', x' \rangle$: what x says in w is true in w' and x identifies themselves as x' in $w'\}$

In (9a), the fact that [LOG] requires x_1 to be bound by λx_1 makes sure that LOGP ends up referring to the attitude holder's recognized self (the 'Logophoric Center'), and this yields obligatory *de se* coreference with the attitude holder. The paraphrase of the resulting meaning is in (10).

(10) $[\![(9a)]\!] \approx$ *In all worlds in which what Kofi says is true, the person Kofi identifies as himself in those worlds marries Afi.* *De se Reading*

The Simplex Binding approach to LOGPs makes a prediction about the possible readings of LOGP with respect to the strict-sloppy ambiguity: it predicts that LOGP cannot have a strict reading in sentences with *only* and in ellipsis. The prediction comes about on standard assumptions about semantic interpretation, namely that bound-variable representations (λ-binding at

[7] There are differences in implementation between Pearson's proposal and the other works cited above, but those differences are immaterial for us in so far as we are merely interested in the Simplex Binding assumption that these accounts are all committed to.

LF) translate to sloppy readings in focus and ellipsis environments (Ross 1967, Partee 1973, Sag 1976, Williams 1977, Reinhart 1983, Heim and Kratzer 1998).

The prediction, however, is not borne out: as we show below, LOGPs in the languages we investigate do admit strict readings (alongside sloppy readings, as expected). The problem has already been highlighted by Culy (1994) and Bimpeh and Sode (2021) for Ewe, and below we provide comparable cross-linguistic data from Ewe, Yoruba and Igbo confirming this conclusion.

3.2 Strict readings across Ewe, Yoruba, and Igbo

Examples (11)-(12) show that in environments involving *only* (association with focus), there is both sloppy and, crucially, strict readings for LOGP. We used a binary acceptability judgment task designed with joint presentation for both strict and sloppy interpretations of the target sentence: speakers were asked to express their acceptability judgments on both paraphrases (one *strict* and one *sloppy*), but they were free to accept as felicitous both sentences, one sentence or none. In each language, the paraphrases were accepted as felicitous interpretations of the target sentence.

(11) *Strict/sloppy readings with 'only' in **Ewe*** (Bassi et al. 2023)

Élì kò yé súsú bé **yè** ɖùdzí lè àwù-dódó ƒé hòʋíʋlí mè.
Eli only FOC think that LOGP win in dress-wear POSS contest inside
'Only Eli thinks that he won the costume contest.'

a. $\rightsquigarrow_{sloppy}$ Eli thinks that he(=Eli) won the costume contest, and Koku doesn't think that he**(=Koku)** won the costume contest, and Kofi doesn't think that he**(=Kofi)** won the costume contest.

b. $\rightsquigarrow_{strict}$ Eli thinks that he(=Eli) won the costume contest, and Koku doesn't think that he**(=Eli)** won the costume contest, and Kofi doesn't think that he**(=Eli)** won the costume contest.

(12) *Strict/sloppy readings with 'only' in **Igbo*** (Bassi et al. 2023)

Náānị́ Ézè chèrè nà **yá** mérìrì nà ásòmpì igósì ákwá.
only Eze think that LOGP win PREP contest show clothes
'Only Ézè thinks that he won the costume contest.'

a. $\rightsquigarrow_{sloppy}$ Eze thinks that he(=Eze) won the costume contest, and Aki doesn't think that he**(=Aki)** won the costume contest, and Ada doesn't think that she**(=Ada)** won the costume contest.

b. $\rightsquigarrow_{strict}$ Eze thinks that he(=Eze) won the costume contest, and Aki doesn't think that he**(=Eze)** won the costume contest, and Ada doesn't think that he**(=Eze)** won the costume contest.

(13) *Strict/sloppy readings with 'only' in **Yoruba*** (Bassi et al. 2023)

Adé nìkan ni ó rò wípé **òun** máa tayọ nínú ìdíje asọ náà.
Ade only FOC RP think that LOGP FUT win inside contest clothes DET
'Only Adé thinks that he will win the costume contest.'

a. $\rightsquigarrow_{sloppy}$ Ade thinks that he(=Ade) will win the costume contest, and Niyi doesn't think that he**(=Niyi)** will win the costume contest, and Ọla doesn't think that she**(=Ọla)** will win the costume contest.

b. $\rightsquigarrow_{strict}$ Ade thinks that he(=Ade) will win the costume contest, and Niyi doesn't think that he**(=Ade)** will win the costume contest, and Ọla doesn't think that he**(=Ade)** will win the costume contest.

Other relevant environments for testing the prediction are ellipsis configurations. To investigate strict and sloppy identity in ellipsis, we did not make use of VP-ellipsis, as this type of ellipsis is not found in the relevant languages. Argument ellipsis is also not an option. As shown in (14), a structure set up to test for strict/sloppy identiy involving *gblɔ̀* 'say' in Ewe requires an overt pronoun.

(14) *No argument ellipsis with 'say' in* ***Ewe***

Élì	gblɔ̀	bé	tsì	lè	dzàdzà-m,	éyé	Édèm	hã	gblɔ̀	*(é).
Eli	say	that	water	be	fall.RED-PROG	and	Edem	too	say	3SG

'Eli said it is raining, and Edem said it, too.'

To our surprise, we also encountered difficulties when testing for stripping configurations, at least for the 'say'/'think'-type verbs. For example, not all of our consultants accepted (15).

(15) *Stripping with 'say' in* ***Ewe***

%Élì	gblɔ̀	bé	tsì	lè	dzàdzà-m,	éyé	Édèm	hã
Eli	say	that	water	be	fall.RED-PROG	and	Edem	too.

'Eli said it is raining, and Edem did, too.'

We did not encounter such difficulties with the verb 'hope', that is, stripping was accepted by all of our consultants for each language. Thus, we demonstrate the availability of strict (and sloppy) readings with logophors embedded under 'hope', which is shown for Ewe in (16), for Igbo in (17), and for Yoruba in (18).

(16) *Strict/sloppy readings with ellipsis in* ***Ewe***

Élì	lè	mɔ́-kpɔ́-m	bé	**yè**	á	ɖè	Àblá.	Yàó hã.
Eli	COP	path-see-PROG	that	LOGP	IRR	marry	Abla.	Yao too.

'Eli hopes that he will marry Abla. Yao does, too.'

a. $\rightsquigarrow_{sloppy}$ Eli hopes that he(**=Eli**) will marry Abla, and Yao also hopes that he(**=Yao**) will marry Abla.

b. $\rightsquigarrow_{strict}$ Eli hopes that he(**=Eli**) will marry Abla, and Yao also hopes that he(**=Eli**) will marry Abla.

(17) *Strict/sloppy readings with ellipsis in* ***Igbo***

Ézè	nwè-rè	òlìlèányá	nà	**yá**	ga	e-mérì	nà	ásòmpì	igósì	ákwá.	Ma
Eze	be-RED	hope	that	LOGP	FUT	PTCP-win	PREP	contest	show	clothes	also

Ada kwa.
Ada too

'Eze is hopeful that he will win the costume contest. Ada is, too.'

a. $\rightsquigarrow_{sloppy}$ Eze hopes that he(=Eze) will win the costume contest, and Ada also hopes that **she(=Ada)** will win the costume contest.

b. $\rightsquigarrow_{strict}$ Eze hopes that he(=Eze) will win the costume contest, and Ada also hopes that **he(=Eze)** will win the costume contest.

(18) *Strict/sloppy readings with ellipsis in **Yoruba***

Adé	ń	rētí	wípé	**òun**	máa	tayọ	nínú	ìdíje	asọ	náà.	Ati	Ọla	pelu.
Ade	PROG	hope	that	LOGP	FUT	win	inside	contest	clothes	DET	and	Ola	also

'Ade hopes that he will win the costume contest and Ọla does, too.'

a. $\rightsquigarrow_{sloppy}$ Ade hopes that he(=Ade) will win the costume contest, and Ọla also hopes that **she(=Ọla)** will win the costume contest.

b. $\rightsquigarrow_{strict}$ Ade hopes that he(=Ade) will win the costume contest, and Ọla also hopes that **he(=Ade)** will win the costume contest.

3.3 The analytical problem

In the previous section we demonstrated the robust availability of strict readings of logophors in focus and ellipsis constructions. The present section lays out the problem this raises for Simplex Binding.

Consider first ellipsis. If ellipsis requires LF/semantic identity ('Parallelism') between the elided phrase and an antecedent phrase (Keenan 1971, Sag 1976, Williams 1977, Tancredi 1992, Rooth 1992a, Fiengo and May 1994, Takahashi and Fox 2005, Merchant 2019, a.o.), then the Ewe sentence in (16), for instance, should be schematically analyzed as in (19) (grey material indicates ellipsis).

(19) *Schematic analysis of (16)*

a. *Antecedent clause*:

Eli hopes [$\boldsymbol{\lambda x_2}$ ... that ***yè***$_{2[\boldsymbol{log}]}$ will marry Abla]

b. *Ellipsis clause*:

Yao hopes [$\boldsymbol{\lambda x_2}$ ... that ***yè***$_{2[\boldsymbol{log}]}$ will marry Abla], too.

Because of the Simplex Binding assumption, the two LOGPs—the overt and the elided one—must be bound by the edge of their respective embedded clauses. This produces a sloppy reading: Yao hopes (*de se*) that he himself will marry Abla. Simplex Binding permits no other representation that both respects the identity condition on ellipsis and can produce a strict reading.

The cases with *only*, for instance in (11), present a similar binding problem. To show this we need to be specific about the analysis of *only* in Ewe, Yoruba and Igbo, but any sensible analysis will do. Concretely, we assume—also for the purpose of preparing the grounds for our proposal later—that these sentences involve the computation of focus alternatives, similar to what is widely assumed for English *only* (e.g. Rooth 1992b). A schematic analysis of (11) is given in (20). The subject *Eli* is marked with the feature FOC(US) which generates alternative structures created by substituting *Eli* with some (relevant) individual. *Only* negates the alternatives.

(20) *Schematic analysis of (11)*

a. *LF*:
Only $\big[$Eli$_{[\text{FOC}]}$ hopes [$\lambda x_{2[log]}$ ... that ***yè***$_{2[log]}$ will marry Abla] $\big]$

b. *Focus Alternatives*:
$\{\big[$Koku hopes [$\lambda x_{2[log]}$ ... that ***yè***$_{2[log]}$ will marry Abla]$\big]$,
$\big[$Kofi hopes [$\lambda x_{2[log]}$ ... that ***yè***$_{2[log]}$ will marry Abla]$\big]$, ... $\}$

With these alternatives the sloppy reading is accounted for: because *yè* is λ-bound, in each of the alternatives the position of *yè* is co-valued (*de se*) with the relevant alternative to Eli. The strict reading, however, requires a different representation, one which appears to be at odds with Simplex Binding, where the value of *yè* would remain constant across all alternatives and pick out Eli.

Something, then, must be changed in the theory. The dilemma we are faced with can be stated as in (21).

(21) LOGP's Dilemma:
If LOGPs have to be bound, how are strict readings possible? If they don't, how to ensure LOGP's obligatory (*de se*) coreference with the attitude holder?

In the next section we turn to our perspective on the dilemma. We will reject Simplex Binding and propose a decompositional analysis of LOGPs which captures the (*de se*) co-reference requirement of LOGPs with a richer structure for LOGPs than so far assumed, one which gives room for strict readings to emerge.

4 Proposal: A decompositional approach to logophoric dependencies

To preview, we propose that the logophoric pronoun (in Ewe, Yoruba and Igbo) underlyingly consists of two syntactic pieces: roughly, **LOGP ≡ [LOG *pro$_i$*]**. *pro$_i$* is a variable over individuals with no binding requirements: it may be free or λ-bound from above. LOG is a presuppositional pronominal feature which roughly equates the reference of *pro$_i$* with the (*self*-counterpart of the) attitude holder, and this produces the *de se*-coreference property of LOGPs. Our proposal also crucially builds on recent ideas in the literature to account for the exceptional behavior of pronominal features in focus and ellipsis environments: their ability to be deactivated when computing focus alternatives and ellipsis identity (e.g. Jacobson 2012, Sauerland 2013). We will show that strict readings of LOGPs are possible if LOG's featural contribution can be suspended in essentially the same way.

If our analysis is correct, then LOGPs are no different from other pronouns at LF in terms of their syntacto-semantic makeup. They contain a variable-part and a semantic feature, just like e.g. the English pronoun *her$_i$* consists of a variable that fixes the reference of the pronoun plus a presuppositional gender feature.

We now turn to the details of the analysis.

4.1 Background: Fake ϕ-features in focus alternatives

A well-known observation in the literature on binding, dating at least as far as Ross (1967), is that pronominal ϕ-features (gender, number, person) have a special status in ellipsis and focus enviroments: their contribution can be ignored across alternatives (see also Heim 2008, Hestvik 1995, among many others). Cases of fake indexicals in focus, as in (22a), as well as fake gender as in (22b), exemplify this:

(22) a. Only I did **my** homework. (Heim 2008, Kratzer 2009)
$\rightsquigarrow$ *bound reading:* No one other than me did **their** own homework.

b. Only Mary did **her** homework.
$\rightsquigarrow$ *bound reading:* No one other than Mary, **male or female**, did their own homework.

To account for such behavior, a prominent approach stipulates something like the following principle (Jacobson 2012, Sauerland 2013, McKillen 2016, Sudo and Spathas 2020, Bruening 2021, Bassi 2021).

(23) **Hypothesis**: The semantic contribution of pronominal features (a presupposition) may be ignored when computing focus alternatives and ellipsis identity.[8]

As previewed, our proposal to capture strict readings of LOGPs relies crucially on this hypothesis. We will propose (section 4.4) that the pronominal feature LOG contained in a LOGP is one whose contribution can be ignored when computing focus alternatives, since it is subject to the hypothesis in (23).[9]

4.2 Setting up a semantics of attitudes with Counterparts

Before going further into our syntax-semantics of LOGPs, we must first lay out some assumption about the general framework. Any theory that incorporates an analysis of reference in attitude contexts must deal with well-known *de re* puzzles that have been discussed ever since the works of Quine (1956) and Kaplan (1968) (cf. also Lewis 1968, Percus and Sauerland 2003, Charlow and Sharvit 2014, Sauerland 2018 and others), and this is what we do in this section. Once we have a working system for modeling reference in (and binding into) attitude environments, we will show how our analysis of LOGP neatly fits in, and how strict readings are accounted for as a virtue of the featural analysis of LOG.

This section may be regarded as somewhat of a long (though necessary) excursus. Readers who are not interested in the details of the assumed framework may wish to skip to section 4.3.[10]

[8] An obvious question is what explains or derives (23) from more prior principles. This is an important question, although one about which we have little to say in this article (cf. Sauerland 2013, Bassi 2021 for different ideas).

[9] Our proposal is most directly inspired by Sauerland's (2013) approach to the phenomenon of strict reading of reflexive pronouns (see Sag 1976, Fiengo and May 1994, Hestvik 1995, McKillen 2016, a.o.). Reflexive anaphors, e.g. English *self*-pronouns in local configurations, permit strict readings in ellipsis and focus environments:

(24) Mary defended **herself** before her lawyer did [~~VP~~].
$\rightsquigarrow$ *strict reading:* ... before her lawyer defended **Mary**.

(25) Only MARY defended **herself**.
$\rightsquigarrow$ *strict reading:* No one else defended **Mary**.

Sauerland derives the strict reading from the idea that reflexive anaphors contain a SELF morpheme with presuppositional semantics and which can be ignored in focus alternatives.

[10] The core proposal in this paper for the syntax and semantics of LOGPs is, we believe, largely orthogonal to the general problem of reference in attitude contexts. It should thus be kept in mind that some of the particular theoretical choices we make in this section 4.2 are done largely for concreteness.

4.2.1 A Lewisian ontology, *de re* puzzles and Counterparts

Our system is loosely based on Sauerland 2018 and Heim 2001, which make use of Lewis 1968's Counterpart ontology to build a compositional analysis of attitude ascriptions. On this ontology, individuals can only occupy one possible world, but an individual can have counterparts in other worlds—more on this below. A first stab at a Lewis-inspired LF for a sentence like *John thinks Mary is a spy* is given in (26). Attitude ascriptions involve quantification over centered-worlds, which are pairs of a world and an individual. In our system, centered-world variables are represented in the structure and saturate argument slots in the denotation of verbal and nominal predicates (see a.o. von Fintel and Heim 2011, Percus 2000). 'w_x' is shorthand for the world-individual pair $< w, x >$.[11]

(26) $\lambda_{w_s^*}$ John thinks$_{w_s^*}$ λ_{w_x} Mary is a spy$_{w_x}$ (to be modified)

In (26) a variable binder over centered-worlds is introduced at the edge of the CPs and binds the centered-world variable on the predicate. Suitable denotations are given in (27a) and (27b) for simple predicates like *spy* and for attitude predicates like *think* (we notate with '`s`' the semantic type of centered-worlds). Throughout, we take individual-denoting phrases (proper names, pronouns) to be interpreted as simple type-*e*, actual-world, individuals: $[\![\text{Mary}]\!]$ = the actual individual Mary.

(27) a. $[\![spy]\!] = \lambda w_x.\lambda z.\ z$ is a spy in w. (*type* $\langle \texttt{s,et} \rangle$)

b. $[\![think]\!] = \lambda w_x \lambda p_{\langle \texttt{s,t} \rangle} \lambda y.\ \forall w'_{x'} \in \text{BEL}_y,\ p(w'_{x'}) = 1.$ (*type* $\langle \texttt{s}, \langle \texttt{st,et} \rangle \rangle$)

The domain of quantification for *think*, BEL, is defined in (28). It is a set of centered-worlds where the world coordinate *w* is as it usually is for belief predicates, and the individual coordinate *x* is a 'recognized self' of the attitude holder.

(28) $\text{BEL}_h \coloneqq \{w_x$: w is compatible with h's beliefs and x is an individual which h identifies as themselves in $w.\}$

Put differently, the *x*-coordinate in (28) is an individual in *w* who, if *h* was put in *x*'s place, *h* would not experience any difference to what they take to actually be the case. We call *x* here a SELF-COUNTERPART of *h* in *w*.

There are two (related) problems with the LF in (26) having to do with the occurrence of *Mary* in the embedded clause. First, since on our Lewis-adopted ontology individuals only occupy one possible world (no trans-world individuals), an interpretation like $[\![(26)]\!] = [\forall w_x \in \text{BEL}_{John},\ Mary\ is\ a\ spy\ in\ w]$ doesn't make much sense; it entails that the predicate *spy*, evaluated in John's belief worlds, applies to the actual Mary, which doesn't inhabit those worlds, and this cannot be satisfied. We need our formalism to rather deliver that the embedded subject denotes not the actual Mary, but different counterparts of hers that live in John's belief worlds. Second, this analysis is not equipped to account for the intuition that attitudes about an individual can be both true and false in so-called 'double-vision scenarios'. These scenarios give rise to certain *de re* puzzles discussed since Quine (1956) and Kaplan (1968). The observation is that sometimes more than one way to specify counterparts of individuals is relevant for the interpretation of such sentences. Imagine that John has been acquainted with a certain Mary in two different ways: he sees her at the beach in a sunny day, and he also sees her trying to

[11] We will discuss the contribution of the matrix abstractor λw_s^* (roughly over actual world-speaker pairs), and its relevance to the grammar of LogPs, in section 5.2. Until then, it will not play much role.

infiltrate into a military base at night. Crucially, John doesn't know that the person at the beach and the infiltrator are in fact one and the same person. John suspects that the infiltrator is a spy, but has no reason to believe that the beach person is a spy. In this double-vision scenario, the sentence *John thinks Mary is a spy* is intuitively both true (by virtue of Mary being the night infiltrator) and false (by virtue of her being the beach person). Double-vision scenarios then point to the need for having a way to semantically encode how exactly attitude holders are acquainted with individuals.

We respond to both of these problems, following Heim (2001), Sauerland (2018), by enriching our representations to include COUNTERPART FUNCTIONS. Individuals indeed occupy only one possible world, but can be systematically linked to their counterparts in other worlds. The exact way they do so depends on some suitable ACQUAINTANCE FUNCTION. The Counterpart technology amounts to a device that maps the evaluation-world individual Mary in (26) into a certain counterpart of Mary that lives in w.

Concretely, we postulate silent counterpart pronouns, $\boldsymbol{C_x^{f^i}}$, which can attach to individual-denoting phrases like *Mary*. An amended LF is in (29).

(29) $\lambda_{w_s^*}$ John thinks$_{w_s^*}$ λ_{w_x}[[$\boldsymbol{C_x^{f^1}}$ Mary] is a spy$_{w_x}$]

Informally, here is how counterpart pronouns work. They consist of two parameters: an indexed acquaintance function f^i, whose content we assume is supplied contextually (by an assignment function), and an individual 'pivot' x which is bound by an attitude holder and is therefore the individual's *self*-counterpart in the relevant worlds. f^i encodes a specific description, say "the woman seen at the beach", by which actual-John is acquainted with actual-Mary. '$\boldsymbol{C_x^{f^i}}$(Mary)' is to be read as *the* f^i*-counterpart of Mary in the world of* x. So, '$\boldsymbol{C_x^{f^2}}$(Mary)', for example, could be a counterpart of Mary in the world of x as she is described in the mind of x as "the woman I saw at the beach"; (29) would thus be true iff John thinks that that woman is a spy.

The Counterpart technology—or its parallels in other systems, notably Concept Generators (Percus and Sauerland 2003, Charlow and Sharvit 2014, a.o.)—allows us to provide an account for double-vision scenarios. A sentence like *John thinks Mary is a spy* uttered in a context where John is acquainted with Mary through two different descriptions as illustrated above could be represented either as in (30a) or (30b)—and one of them could be true while the other false. The difference is located in the different acquaintance functions, which determine different individuals as counterparts of the same actual Mary.

(30) a. $\lambda_{w_s^*}$ John thinks$_{w_s^*}$ λ_{w_x}[[$\boldsymbol{C_x^{f^1}}$ Mary] is a spy$_{w_x}$]
E.g. John thinks: "the person I saw trying to infiltrate the base is a spy".

b. $\lambda_{w_s^*}$ John thinks$_{w_s^*}$ λ_{w_x}[[$\boldsymbol{C_x^{f^2}}$ Mary] is a spy$_{w_x}$]
E.g. John thinks: "The person I saw at the beach is a spy".

The next section provides more content and definitions necessary for the formal details of how counterpart operators are interpreted (but readers not interested in the details can skip to section 4.3). Before that, we give in (31) the LF for a sentence with a free pronoun instead of a proper name in the embedded clause. The interpretation procedure is just like it is for (29), except that the value of the embedded subject she_i is supplied, as usual, by an assignment function. A Counterpart function applies to it to turn it into the suitable individuals that reside in the belief worlds of John.

(31) a. John thinks that she is a spy.

b. $\lambda_{w_s^*}$ John thinks$_{w_s^*}$ λ_{w_x}[[$C_x^{f^1}$ she$_j$] is a spy$_{w_x}$]

4.2.2 Counterparts: formal details

Let us first define acquaintance functions. These, adopting the spirit of Sauerland (2018)'s suggestion, are functions that relate an individual to another by some suitable description from a first-person perspective.

(32) **Acquaintance Function**. A function f^δ of type $\langle \texttt{e,e} \rangle$ mapping individuals to their world-mates is an *Acquaintance Function* iff there is a definite description of individuals, δ, containing a first-person pronoun such that for any $x \in dom(f^\delta)$, $f^\delta(x) =$ the individual in the world of x who x knows as 'δ'.

Take for instance the 1st-person-pronoun-containing description δ='my math teacher'. Many individuals are acquainted with many other (world-mate) individuals through this description. When John utters it, it refers to John's math teacher. The acquaintance function that corresponds to it, defined in (33), maps any individual x for which the description is defined (i.e. any x who knows someone as their math teacher), to the individual who fits this description for x.

(33) $f^{\text{my math teacher}} = \lambda x.$ the person in x's world who x knows as x's math teacher

The special case of BELIEF-BASED acquaintance functions is defined in (34): they guarantee a value for any *self*-counterpart of the individuals in their domain.

(34) **Belief-based Acquaintance Function**. An Acquaintance Function f is *belief-based* iff $\forall h \in dom(f)[\forall w_x \in \text{BEL}_h[x \in dom(f)]]$.

Belief-based acquaintance functions are those that are needed for attitude contexts. Informally, an acquaintance function f is belief-based if it preserves the underlying acquaintance relationship through which perspective-holders are acquainted with acquaintants, for all of their *self*-counterparts. Where x is a *self*-ounterpart of some attitude holder h, $f(x)$ is a world-mate of x and is the individual that x is acquainted with via the same description that specifies how h is acquainted with $f(h)$.

Let us suppose further that *self*-counterparts of individuals have a unique actual SOURCE in the following sense: every possible individual is the self-counterpart of at most one actual individual. Thus, where h and j are distinct actual individuals, there is no world-individual pair $< w,x >$ which is in both BEL_h and BEL_j (see definition of BEL in (28)).[12] With this, we can explicate the semantics of Counterpart operators in (35).

(35) **Counterpart**. $[\![C_x^{f^i}]\!]^g(y)$ ('the f^i-Counterpart of y for x') is defined iff $g(f^i)$ is a belief-based acquaintance function and $\big(g(f^i)\big)(h){=}y$, where h is the Source of x (i.e. h is the unique individual that satisfies $w_x \in \text{BEL}_h$, for some w).
If defined, $[\![C_x^{f^i}]\!](y) = \big(g(f^i)\big)(x)$.

[12]This assumption is defensible in so far as we hold that no two individuals have identical experiences and completely compatible beliefs about the world. It thus ignores metaphysical edge cases such as the two-gods scenario discussed in Lewis (1979: p.520).

$[\![C_x^{f^i}]\!]^g(y)$ is the individual in x's world who x is mapped to ('acquainted with') via the same acquaintance function which maps x's Source (the attitude holder) to y. Note that for $[\![C_x^{f^i}]\!]^g(y)$ to be defined, y needs to live in the world of the Source of x; in the LFs in (30), for instance, y is the actual Mary and she lives in the world of the actual attitude holder John, who is the Source of his self-counterpart x. The interpretation of (36) is in (36).

(36) $\lambda_{w_s^*}$ John thinks$_{w_s^*}$ λ_{w_x}[[$C_x^{f^1}$ Mary] is a spy$_{w_x}$] (repeated from (29))

(37) $[\![36]\!] = \forall w_x$, w is compatible with John's beliefs and x is a *self*-counterpart of John in w: the f^i-counterpart of Mary for x is a spy in w.

Counterparts are f-dependent (description-dependent), so we cannot generally talk simply about *the* counterpart of y in w. As discussed earlier, the truth of attitude ascriptions is sometimes sensitive to the way individuals become known to attitude holders. Within our system, the same individual y can have different counterparts in one and the same doxastically-accessible world, so $C_x^{f^1}(y) \not\equiv C_x^{f^2}(y)$ when f^1 and f^2 are two different belief-based acquaintance functions. This solves the *de re* puzzles described at the end of the previous section:

4.3 The meaning of LOGP and a derivation of basic sentences

The previous section provided us with tools that are rich enough to properly handle reference in attitude contexts. Now we can go back to our innovation regarding LogPs. Consider again the example in (38).

(38) Élì súsú bé **yè** ɖùdzí. **Ewe**
Eli think that **LOGP** win
'Eli$_1$ thinks that he$_1^{de\text{-}se}$ won.'

As previewed, we propose that the logophoric pronoun (in Ewe, Yoruba and Igbo) underlyingly consists of two semantically active syntactic pieces: roughly, ignoring counterpart functions for a moment, **LOGP ≡ [LOG$_x$ [*pro$_i$*]]**. *pro$_i$* is a simple individual variable with no binding requirement: it may be free or bound, and if free its value is supplied by an assignment function. LOG$_x$is an indexed presuppositional feature, whose index must be bound by a centered-world abstractor at the edge of a clause, and is thus equated with (a counterpart of) the attitude holder.[13]

(39) A Logical Form (first pass): $\lambda_{w_s^*}$ Eli thinks$_{w_s^*}$ that λ_{w_x} $\left[\underbrace{[\text{LOG}_x\ pro_i]}_{\text{LOGP}} \text{won}_{w_x} \right]$

[13]We are deliberately saying that the index on LOG$_x$ must be bound at the edge of a **clause**, not necessarily at the edge of a **complement** clause as is usually assumed in accounts of logophoric pronouns (von Stechow 2004, Pearson 2015, a.o.) The reason for this more relaxed requirement will be discussed in section 5.2, where we show that it helps explain a further distributional property of LOGPs.

The meaning of $\textsc{log}_x$ is given in (41). $\textsc{log}_x$ functions like other pronominal features in being a pure-presupposition trigger (a partial identity function), acting as a filter on the possible values of its argument. Specifically, it imposes the condition that its argument's value is identified with the index on LOG.[14]

(41) $[\![\textsc{log}_x]\!] = [\lambda z : z = x.\ z]$[15]

Remember that given our ontology, if pro_i's value is an actual-world individual—a possibility which we need to assume for the purpose of deriving strict readings—then its occurrence in an attitude context in (39) means that it must be 'turned into' a counterpart of that individual that lives in the embedded world. Officially, then, we replace (39) with (42), which, just like we had in (31), appends a counterpart function to pro_i (but differently from (31), also has $\textsc{log}_x$). Consequently, the interpretation of (31) is given in (43).

(42) <u>A Logical Form (amended)</u>:
$\lambda_{w_s^*}$ Eli thinks$_{w_s^*}$ that λ_{w_x} $\big[$ [$\textsc{log}_x$ $\boldsymbol{C_x^{f^1}}$ (pro_i)] won$_{w_x}\big]$

(43) $[\![\textsc{LogP}]\!]^g = [\![\textsc{log}_x\ [\boldsymbol{C_x^f}\ pro_i]]\!]^g$ (for any belief-based f)
$= \ :\ \underbrace{[\![C_x^f\,(pro_i)]\!]^g = x}_{\textsc{log}_x\text{'s presupposition}}\ .\ [\![C_x^f\,(pro_i)]\!]^g$

$\textsc{log}_x$ thus forces the relevant counterpart of pro_i to be identical to the self-counterpart of the attitude holder. This is as desired, as it captures the *de se* reading of LOGP: it ensures that the value of the whole LOGP is the *self*-counterpart of the attitude holder (no matter how the

[14]A reviewer asks in what sense the LOG feature can be counted as a pronominal feature. Semantically, as we lay out in the next section, our denotation for LOG adds a presuppositional restriction on the value of a variable – in that sense it matches what has been proposed about the semantics of ϕ-features. And syntactically, there is evidence that logophors trigger dedicated logophoric agreement on the verb in languages like Ibibio, shown for the logophor *imọ* in (40a). Hence, the LOG feature behaves morpho-syntactically like other ϕ-features in this respect.

(40) *Logophoric agreement in Ibibio* (Newkirk 2019)

a. Ekpe$_i$ a-bo ke **imọ**$_i$ **ì**-ma **í**-to Udo.
Ekpe 3SG-say C **LOG** **LOG**-PST **LOG**-hit Udo
'Ekpe says that he hit Udo.'

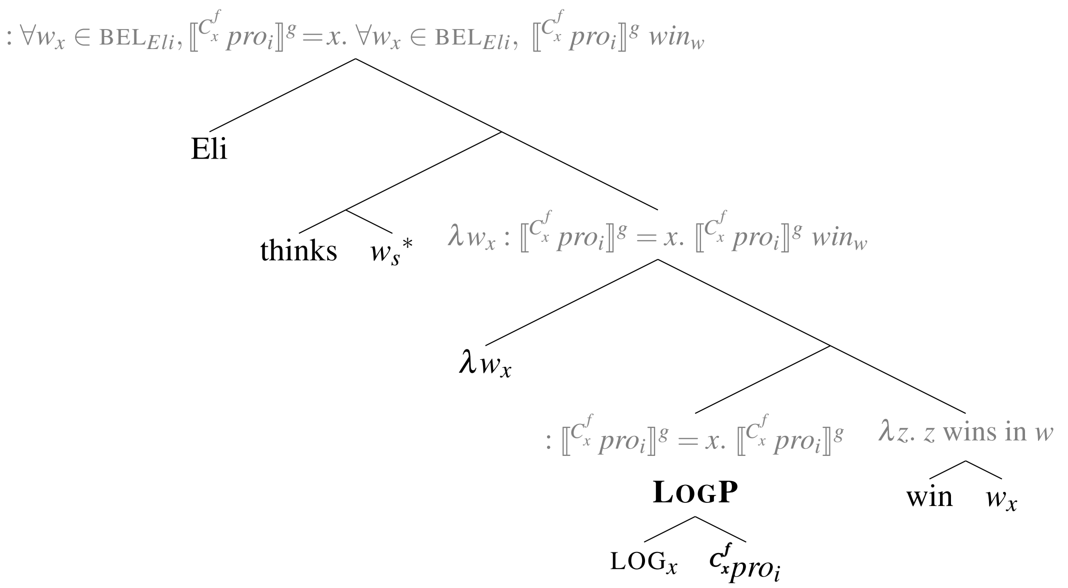
b. Ekpe$_i$ a-bo ke **anye**$_{i/k}$ **a**-diyọn̄ọ ikwo ikwo mfọnmfọn.
Ekpe 3SG-say C **3SG** **3SG**-know sing song well
'Ekpe says that he sings well.'

While this property cannot be investigated in the languages which we focus on in this paper since Ewe, Yoruba and Igbo do not display subject verb agreement, we note that in the languages which do display subject verb agreement and logophors, dedicated inflectional morphology is attested, as one would expect from a pronominal feature.

[15]We employ the notation of Heim and Kratzer (1998) for encoding partial functions, where the part between a colon and a dot defines the domain of the function (and where partiality is meant to model the presuppositional dimension).

counterpart of pro_i is fixed, i.e., no matter the specific value of f). Here's a complete semantic derivation of (38), for an arbitrary f:[16]

(45)



(46) $[\![45]\!]$ =
Presupposition (simplified): In all w_x where w is compatible with Eli's beliefs and x is a *self*-counterpart of Eli in w: x is the f-counterpart of $[\![pro_i]\!]^g$ for x.
Assertion: In all such w_x: the f-counterpart of $[\![pro_i]\!]^g$ for x wins in w.

What can $[\![pro_i]\!]^g$ be? Apart from contextual recoverability, free pronouns are not normally referentially constrained. But here, pro_i's value cannot be just anyone contextually salient; if it is an actual-world individual, it must be the (actual) attitude holder—Eli. This is because only in that case will the presupposition in (46) generated by LOG_x be satisfied. To wit, this presupposition equates x, a *self*-counterpart of Eli, with a counterpart of $[\![pro_i]\!]^g$. By definition of Counterpart in (35), possible individuals have only one Source, which means that the Source of x must be the Source of $[\![pro_i]\!]^g$ as well. From these two, and from the fact that Eli is the Source of x, it follows that $[\![pro_i]\!]^g = \text{Eli}$. Therefore we can safely replace (46) with (47):

(47) $[\![45]\!]$ =
Presupposition: In all w_x such that w is compatible with Eli's beliefs and x is a *self*-counterpart of Eli in w: x is the f-counterpart of **Eli** for x.
Assertion: In all such w_x: the f-counterpart of **Eli** for x wins in w.

[16]Given that we now represent presuppositions, we replace the entry for *think* we had in (27b) with the one in (44) that encodes how presupposition are projected, adopted after Karttunen 1974 and Heim 1992. This entry is used in the calculation in (45).

(44) $[\![\text{think}_{w_s{}^*}]\!]^g = \lambda p_{\langle s,t\rangle} \lambda y : \forall w_x \in \text{BEL}_y,\ p(w_x) \text{ is defined.}$
$\forall w_x \in \text{BEL}_y,\ p(w_x) = 1$

Similar lexical entries need to be written for predicates of speech, desire, hopes etc. All that matters for our purposes is that for each such predicate the domain of quantification is a set of centered-worlds, and that the presupposition of LOG projects to BEL.

This faithfully represents the *de se* construal of LOGPs. Now what can the value of the acquaintance function-variable f be? It too, in principle, merely needs to be recoverable from context (by some salient-enough description); but here too, f is semantically restricted: only those values are possible which result in a presuppositional statement that can be safely accommodated. In our case, because $[\![pro_i]\!]^g$ must be Eli, f must in turn be an acquaintance function which maps Eli to himself both in the actual world and in his alternative worlds. The option in (48a), which is just the identity function (underlined by the *se* description, "me"), will always be possible here; also the one in (48b) is ok in natural contexts in which Eli knows himself as the person called 'Eli'. Some possibilities are out in most natural contexts since they would incur a presupposition failure, e.g. (48c) (if Eli $\neq$ Ann); and others are heavily context-dependent, for instance (48d) which would satisfy the presupposition only in contexts where Eli wears a red costume and identifies himself as such.[17] This nicety in the options for f will become relevant soon.

(48) *Options for the value of* f *in (46)*

a. ✓$f^{\text{I/me}} = [\lambda x.\ x]$

b. ✓$f^{\text{The person I know as 'Eli'}} = [\lambda x.$ the person who x knows as 'Eli'$]$

c. ✗$f^{\text{The person I know as 'Ann'}} = [\lambda x.$ the person who x knows as 'Ann'$]$

d. ? $f^{\text{The person I know as wearing a red costume}} = [\lambda x.$ the person who x knows as wearing a red costume$]$

4.4 Strict readings by ignoring LOG from alternatives

We have just shown how to derive obligatory *de se* readings by splitting up LOGPs and by assuming that the LOG feature introduces a presupposition (without any formal binding dependency between pro_i and the matrix subject). By this we merely replicated a basic result already obtained by Simplex Binding, just in a more complicated way. But the complication allows us to derive strict readings, to which we now turn. Consider again an example that brings strict-sloppy ambiguity to light. We repeat example (11) in an abbreviated form in (49), with which we will exemplify our analysis.

(49) Élì kò yé súsú bé **yè** ɖùdzí. **Ewe**
Eli only FOC think that LOGP win
'Only Eli thinks that he won.'

a. $\rightsquigarrow_{sloppy}$ No one$_j$ but Eli thinks **they**$_j$ won.

b. $\rightsquigarrow_{strict}$ No one but Eli$_i$ thinks **he**$_i$(=Eli) won.

We continue to analyze these constructions as involving computation of focus alternatives triggered by a focus feature on the subject, as represented in (50a). For concreteness we take alternatives to be syntactic objects, LFs (Fox and Katzir 2011). *only* says that the prejacent (its

[17] Acquaintance functions which don't map Eli to himself in the actual world (i.e., acquaintance functions which are not *reliable* in the sense of Pearson 2015) would not result in a satisfied presupposition in (45), even if Eli mistakenly associates himself with that description. Imagine for instance that Eli believes he's the person wearing the red costume, but is mistaken about it (he actually wears a different costume). In that context (48d) cannot be the value for f given LF (45).

sister) is true and all the alternatives are false.[18] Now if LOG contributes its meaning across the focus alternatives, as represented in (50b), we are still short of deriving the intended strict reading.

(50) a. LF: Only $\big[$Eli$_{[\textbf{FOC}]}$ thinks λw_x [[$_{\text{LogP}}$ $\textbf{[LOG}_{\boldsymbol{x}}$ $\textbf{[}C^f_x$ $\boldsymbol{pro_i}\textbf{]]}$] won$_{w_x}$] $\big]$

b. Alt's: $\big\{$ Kofi thinks λw_x [[$_{\text{LogP}}$ $\textbf{[LOG}_{\boldsymbol{x}}$ $\textbf{[}C^f_x$ $\boldsymbol{pro_i}\textbf{]]}$] won$_{w_x}$] ,
Koku thinks λw_x [[$_{\text{LogP}}$ $\textbf{[LOG}_{\boldsymbol{x}}$ $\textbf{[}C^f_x$ $\boldsymbol{pro_i}\textbf{]]}$] won$_{w_x}$] , ...$\big\}$

In particular, (50) fails at the level of focus alternatives. To appreciate why, notice first that since pro_i refers to Eli, which as we have shown above is forced on us by the presence of LOG at the prejacent, then pro_i's reference to Eli will remain constant across the alternatives. But then, by the same reasoning, LOG's presence in the alternatives results in the false information that the relevant alternatives to Eli (Kofi, Koku) are in actuality equated with Eli, which cannot be.[19]

However, we assume as discussed in section 4.1 that LOG—being a pronominal semantic feature—is subject to the hypothesis in (23) and thus can be ignored when computing focus alternatives, like fake ϕ-features on bound pronouns. We implement the idea by letting LOG be deleted from the tier of alternatives (though not from the prejacent). The relevant derivation is in (51).

(51) *Analysis of (49) with* LOG *deleted from alternatives*[20]

a. LF: Only $\big[$Eli$_{[\textbf{FOC}]}$ thinks λw_x [[$_{\text{LogP}}$ $\textbf{[LOG}_{\boldsymbol{x}}$ $\textbf{[}C^f_x$ $\boldsymbol{pro_i}\textbf{]]}$] won$_{w_x}$] $\big]$

b. Alt's: $\big\{$ Kofi thinks λw_x [[$_{\text{LogP}}$ $\textbf{[}\cancel{\textbf{LOG}_{\boldsymbol{x}}}$ $\textbf{[}C^f_x$ $\boldsymbol{pro_i}\textbf{]]}$] won$_{w_x}$] ,
Koku thinks λw_x [[$_{\text{LogP}}$ $\textbf{[}\cancel{\textbf{LOG}_{\boldsymbol{x}}}$ $\textbf{[}C^f_x$ $\boldsymbol{pro_i}\textbf{]]}$] won$_{w_x}$] , ...$\big\}$

As above, the referent of pro_i remains Eli across the alternatives. But since LOG is active only in the prejacent now, the offending presupposition is absent in the alternatives. The interpretation of this configuration is given in (52), and *only* negates the alternatives.

(52) *The Interpretation of the prejacent and alternatives in (51)*

a. *Prejacent*:
$$:\underbrace{\forall w_x \in \text{BEL}_{Eli}, [\![C^f_x\, Eli]\!]^g = x}_{presupposition}.\quad \forall w_x \in \text{BEL}_{Eli},\ [\![C^f_x\, Eli]\!]^g\ win_w$$

b. *Alternatives*:
$$\big\{\ \forall w_x \in \text{BEL}_{Koku},\ [\![C^f_x\, Eli]\!]^g\ win_w\ ,\quad \forall w_x \in \text{BEL}_{Kofi},\ [\![C^f_x\, Eli]\!]^g\ win_w\ , ...\big\}$$

[18]We evidently assume in (50a) that *only* scopes at LF over the whole rest of the clause. This is done for simplicity's sake. We could instead adopt the bi-partite structure *[[only DP$_{Foc}$] VP]* (Wagner 2006, a.o.), wher *only* forms a constituent with its focused associate (the subject). Our main proposal isn't affected by this choice, as long as the subject DP formally triggers focus alternatives.

[19](50b) also results in the information that the alternatives Kofi, Koku identify themselves in their mind with Eli (this part by itself is not contradictory).

[20]The deletion operation is adopted to simplify the presentation. Instead of syntactically deleting LOG, we could neutralize its contribution by postulating (as in Sauerland 2013) that the meaning of a semantic feature can be reset in the alternatives to the total-identity function [$\lambda f.f$]. Both implementations capture the idea the LOG's contribution is ignored in alternatives.

The specific interpretation of this depends, of course, on the specific acquaintance function chosen for f. All that is required is for f to represent an acquaintance function that links Eli as well as all of his alternatives (Koku, Kofi,...) to Eli, and that furthermore they all mentally associate with Eli. If, for example, it is common ground that everyone knows Eli by the name 'Eli', then plugging in for f the value in (48b), will result in a suitable strict reading.

This proposal makes the correct prediction that the strict reading can be achieved via many different ways of specifying f. Koku and Kofi do not have to be acquainted with the referent of LOGP through the description 'the person called Eli'; they only need to know him through some shared description, if one can be accommodated. That the prediction is borne out is exemplified in (53). In this scenario, Koku and Kofi are acquainted with a certain man in a red costume, but do not know it is Eli. We call it a STRICT-UNKNOWN scenario.

(53) *Context for strict unknown identity (**Ewe**):*
There is a costume contest. Eli, a participant who was wearing a red costume, overhears the judges of the contest debating, and concludes from what he hears that he is going to be declared as the winner. Koku and Kofi, who watched the costume show, are wrong about the identity of the man with the red costume; they don't know it was Eli. (They might even disagree among themselves who it was.) But they don't think that he, whoever he is, will win.

Élì kò yé súsú bé **yè** ɖùdzí lè àwù-dódó ʄé hòʊíʊlí mè.
Eli only FOC think that LOGP win in dress-wear POSS contest inside
'Only Eli thinks that he won the costume contest.'

Our Ewe consultants judged the sentence felicitous and true in the scenario. The sentence entails that Koku and Kofi don't think that the man in the red costume won—despite their lack of awareness that it is Eli.

The facts hold in Igbo and Yoruba as well. In Strict-Unknown scenarios, LOGP *yá* in Igbo and LOGP *òun* in Yoruba are felicitous, see (54) and (55).

(54) *Context for strict unknown identity (**Igbo**):*
There is a costume contest. Eze, a participant who was wearing a red costume, overhears the judges of the contest debating, and concludes from what he hears that he is going to be declared as the winner. Aki and Ada, the other contestants, are wrong about the identity of the man with the red costume; they don't know it was Eze. (They might even disagree among themselves who it was.) But they don't think that he, whoever he is, will win.

Náānị Ézè chèrè nà **yá** mérìrì nà ásòmpì igósì ákwá.
only Eze think that LOGP win PREP contest show clothes
'Only Eze thinks that he won the costume contest.'

(55) <u>*Context for strict unknown identity (**Yoruba**):*</u>
There is a costume contest. Ade, a participant who was wearing a red costume, overhears the judges of the contest debating, and concludes from what he hears that he is going to be declared as the winner. Niyi and Ọla, who are also contestants in the costume show, are wrong about the identity of the man with the red costume; they don't know it was Ade. (They might even disagree among themselves who it was.) But they don't think that he, whoever he is, will win.

Adé	nìkan	ni	ó	rò	wípé	**òun**	máa	tayọ	nínú	ìdíje	asọ	náà.
Adé	only	FOC	RP	think	that	LOGP	FUT	win	inside	contest	clothes	DET

'Only Ade thinks that he will win the costume contest.'

The reason LOGPs are predicted to be licensed in strict-unknown cases is that the value for f in our LF configuration in (51) can be resolved to the description in (48d). This description is salient in the context, and it refers to Eli in Eli's mind but not to who Koku and Kofi associate with the description 'Eli'.[21]

As one of our consultants emphasized, if Eli does not know that he is the man in the red costume, the target sentence in (53) is not felicitous in such strict-unknown scenarios. Thus, the attitude holder in the prejacent (*Eli*) still has to be familiar with himself as the man in the red costume. This is expected for the same reason that *de re* readings of LOGPs are out in basic sentences (cf. section 2.2): LOG's presupposition, which in the prejacent cannot be ignored, imposes the *de se* condition on the use of LOGP. This concludes our basic proposal of the strict reading of LOGPs.

4.5 Sloppy readings

As for the sloppy reading of LOGPs, we can derive it by binding the variable-part (pro_i) of LOGP directly to the matrix subject, as in the representation in (56), and setting the value of f to be the identity function $[\lambda x.\ x]$ (underlined by the *se* description "me"; see (48a)). Since the binding dependency persists across the alternatives, the value of pro_i co-varies in alternatives with the respective attitude holder. The sloppy reading thereby obtains (in fact, whether or not LOG gets suspended in alternatives).

(56) *Sloppy reading through binding* pro$_i$ *and with* f = $[\lambda x.\ x]$

a. <u>LF</u>: Only $\big[$Eli$_{[\mathbf{FOC}]}$ $\boldsymbol{\lambda_i}$ $[\boldsymbol{t_i}$ thinks λw_x [[$_{\text{LogP}}$ [LOG$_x$ [$\boldsymbol{C^f_x}$ pro_i]]] won$_{w_x}$] $\big]$

b. <u>Alt's</u>: $\big\{$ Kofi $\boldsymbol{\lambda_i}$ $[\boldsymbol{t_i}$ thinks λw_x [[$_{\text{LogP}}$ [$\cancel{\text{LOG}_x}$ [$\boldsymbol{C^f_x}$ pro_i]]] won$_{w_x}$] $\big]$,
Koku $\boldsymbol{\lambda_i}$ $[\boldsymbol{t_i}$ thinks λw_x [[$_{\text{LogP}}$ [$\cancel{\text{LOG}_x}$ [$\boldsymbol{C^f_x}$ pro_i]] won$_{w_x}$] $\big]$ ...$\big\}$

[21] What if a context furnishes no suitable definite description that could be the value of f in this derivation? Our predictions might change. Imagine we cook up a context where it is impossible to find any (salient) description which links Eli and of all his relevant alternatives to Eli and is vivid enough in their minds (i.e., across the doxastically-accessible worlds) as such. In such a context, we predict that sentence (49) would not support the strict reading (we thank Amir Anvari (p.c.) for raising a similar point to us). We think, however, that finding convincing cases of such contexts is not trivial, as there is arguably always *some* shared description that can be accommodated whose reference in the minds of the relevant individuals would be the matrix subject. In the absence of independently justifiable constraints on what concepts can be accommodated in a context, it is not entirely clear to us how to create the relevant scenarios. We leave this as an open question for now.

4.6 Strict Ellipsis

We have applied our analysis to sentences with focus and *only*, though earlier we saw that strict readings for LOGPs show up in ellipsis constructions as well. Recall e.g. (57), repeated from (16).

(57) Élì lè mɔ́-kpɔ́-m bé **yè** á ɖè Àblá. Yàó hã. **Ewe**
Eli COP path-see-PROG that LOGP IRR marry Abla. Yao too.
'Eli hopes that he(=**Eli**) will marry Abla. **Yao** does too.'

a. $\rightsquigarrow_{sloppy}$ Yao$_j$ hopes that **Yao**$_j$ will marry Abla.

b. $\rightsquigarrow_{strict}$ Yao$_j$ hopes that **Eli**$_i$ will marry Abla.

Ellipsis licensing requires an antecedent with a parallel meaning. We follow Tancredi (1992), Rooth (1992a), Takahashi and Fox (2005), Merchant (2019) and many others in linking the Parallelism requirement to the theory of focus:

(58) **Parallelism requirement on ellipsis**: To license ellipsis of some phrase, the phrase must be contained in a sentence E whose focus alternatives have a member E' such that $[\![E']\!] = [\![\text{ANT}]\!]$,
where ANT is a sentence uttered in the nearby discourse.

The special status of LOG as a pronominal feature in our theory helps with the strict reading in ellipsis, too. We will adopt a version of Sauerland 2013's proposal on alternatives to work for ellipsis. We stipulate that presuppositions coming from pronominal features in an antecedent ANT may be ignored for the purpose of computing the identity statement for ellipsis. Thus, we replace (58) with (59):

(59) **Parallelism for ellipsis, 2nd version**: To license ellipsis of some phrase, the phrase must be contained in a sentence E whose focus alternatives have a member E' such that $[\![E']\!] = [\![\textbf{ANT*}]\!]$,
where ANT* is just like the uttered ANT except that pronominal features in ANT may be removed from ANT*.

Parallelism is satisfied if what is elided is not a LOGP but the more deficient ORDP: a pronoun with the same referential core *pro$_i$* but without the LOG feature (below we abstract away from counterpart operators, to simplify the representations and because they are irrelevant to the point).

(60) a. Antecedent clause LF:
Eli hopes λw_x [[$_{\text{LOGP}}$ **[LOG *pro$_i$*]**] marries$_{w_x}$ Abla]

b. Ellipsis clause LF:
Yao$_{[\textbf{FOC}]}$ hopes λw_x [[$_{\text{ORDP}}$ **[** ***pro$_i$*]**] marries$_{w_x}$ Abla]

The two clauses satisfy our version of the Parallelism requirement, as follows. One of the focus alternatives of the Ellipsis clause in (60b) is the one where *Yao* is replaced by *Eli*, in (61).

(61) *A member of the focus alternatives of (60b)*:
Eli hopes λw_x [[$_{\text{ORDP}}$ **[** ***pro$_i$*]**] marries$_{w_x}$ Abla]

Indeed, (61) is derived from (60a) by removing the feature LOG, and therefore it is an appropriate ANT*. Ellipsis is consequently licensed by condition (59).

Like before, because we have a LOGP in ANT, the value of pro_i is restricted by the LOG feature to be Eli. The ellipsis clause must then contain an occurrence of the same variable in that position (the underlying acquaintance function f could be, for example, "the person called 'Eli'", assuming both Eli and Yao associate this person with Eli).

5 Further consequences of the account

This section explores consequences of our theory regarding two other properties of logophors: their ability to take long-distance antecedents in multiple attitude sentences (section 5.1), and their complementary distribution with the 1st-person pronoun (section 5.2). The account we offer in section 5.2 captures an observation which, as far as we know, has not received adequate attention (let alone an explanation) in the formal literature, namely why LOGPs cannot be anteceded by 1st-person attitude holders (Hagège 1974, Hyman and Comrie 1981).

5.1 Long-distance antecedents

In this section we analyze the ability of logophors to take long-distance antecedents. As we show in (62)-(64), LOGPs embedded under more than one attitude predicate can co-refer with either the local attitude holder or the more distant attitude holder. Overall, the co-reference patterns are stable across attitude predicates and languages.[22] Our observations are in line with what has been reported for Ewe and Yoruba in previous literature (Clements 1975, Manfredi 1987, Pearson 2015, Anand 2006). We further contribute multiple embedding data from Igbo, which patterns with Yoruba and Ewe.

(62) *Multiple embeddings in* ***Ewe***

a. Kòkú$_1$ súsú bé Kòfí$_2$ bé **yè**$_{1/2}$ dè Àfí.
Koku think that Kofi say LOGP marry Afi
'Koku thinks that Kofi said that he married Afi.'

b. Kòkú$_1$ súsú bé Kòfí$_2$ dʒí bé **yè**$_{1/2}$ á dè Àfí.
Koku think that Kofi want that LOGP IRR marry Afi
'Koku thinks that Kofi wants to marry Afi.'

c. Kòkú$_1$ súsú bé Kòfí$_2$ le mɔ́-kpɔ́-m bé **yè**$_{1/2}$ á dè Àfí.
Koku think that Kofi COP path-see-PROG that LOGP IRR marry Afi
'Koku thinks that Kofi hopes that he will marry Afi.'

[22] There was one exception. For the multiple embedding including 'want' in Igbo (64b), one of our consultants did not accept the co-reference reading with the long distance antecedent.

(63) *Multiple embeddings in **Yoruba***[23]

a. Adé$_1$ rò wípé Olú$_2$ sọ wípé **òun**$_{1/2}$ nifẹ́ Ọlá.
Ade think that Olu say that LOGP love Ola
'Ade thinks that Olu said that he loves Ola.'

b. Adé$_1$ rò wípé Olú$_2$ fẹ́ pékí **òun**$_{1/2}$ máa fẹ́ Ọlá.
Ade think that Olu want that LOGP FUT marry Ola
'Ade thinks that Olu wants to marry Ola.'

c. Adé$_1$ rò wípé Olú$_2$ ń rètí wípé **òun**$_{1/2}$ máa fẹ́ Ọlá.
Ade think that Olu PROG hope that LOGP FUT marry Ola
'Ade thinks that Olu hopes that he will marry Ola.'

(64) *Multiple embeddings in **Igbo***

a. Ézè$_1$ chèrè nà Úchè$_2$ sị̀ nà **yá**$_{1/2}$ hụ̀rụ̀ Àdá n'ányá.
Eze think that Uche say that LOGP see Ada PREP=eye
'Eze thinks that Uche said that he loves Ada.'

b. Ézè$_1$ chèrè nà Úchè$_2$ chọ̀rọ̀ ka **yá**$_{\%1/2}$ lu-o Àdá.
Eze think that Uche want that LOGP marry-FUT Ada
'Eze thinks that Uche wants to marry Ada.'

c. Ézè$_1$ chèrè nà Úchè$_2$ nwè-rè olìlèányá nà **yá**$_{1/2}$ gà à-lú Àdá.
Eze think that Uche be-RED hope that LOGP FUT PTCP-marry Ada
'Eze thinks that Uche is hopeful that he will marry Ada.'

We have the tools to account for the option of LOGPs to take long-distance antecedents: both attitude predicates introduce complement clauses and therefore abstraction over centered-worlds, so in principle either can bind LOGP. More specifically, either can bind the index on LOG. Taking sentence (62b) as representative, and leaving out counterpart operators for a moment, we assume that it can correspond either to representation (65a) or to representation (65b), where the difference is underlined.

(65) a. Koku thinks $\lambda_{\underline{w_x}}$ Kofi wants $\lambda_{w'_{x'}}$ $\big[\,[\,\text{LOG}_{\underline{x}}\ pro_i]\ \text{marry}_{w'_{x'}}\ \text{Afi}\big]$

b. Koku thinks λ_{w_x} Kofi wants $\lambda_{\underline{w'_{x'}}}$ $\big[\,[\,\text{LOG}_{\underline{x'}}\ pro_i]\ \text{marry}_{w'_{x'}}\ \text{Afi}\big]$

But not quite; once again, our background ontology and semantics requires the representation to be decorated with counterpart operators. We will disregard any counterpart operators that in both LFs need to be inserted anyway on the individual-denoting phrases *Kofi* and *Afi*, and restrict attention to those that need to be inserted inside LOGP. For (65b), which is intended to capture the local-binding option, a corresponding counterpart operator with the same index as on LOG is added, so that pro_i ends up referring to a counterpart of the embedded subject (Kofi); this is shown in (66b). For (65a), which is intended to capture the long-distance reading, a suitable operator is analogously added on pro_i, and in addition the whole LOGP is appended with a

[23]Note that in (63b) the complementizer changes in the most embedded clause.

further counterpart operator bound by the embedded subject, so that LOGP as a whole denotes a counterpart of (the counterpart of) the matrix subject Koku, that resides in the embedded-bound worlds w'. This is shown in (66a).[24]

(66) a. Koku thinks $\lambda_{\underline{w_x}}$ Kofi wants $\lambda_{w'_{x'}}$ $\big[\, \boldsymbol{C}^{\boldsymbol{f'}}_{\boldsymbol{x'}}\, [\, \text{LOG}_{\underline{x}}\, [\boldsymbol{C}^{\boldsymbol{f}}_{\underline{\boldsymbol{x}}}\ (pro_i)]]\ \text{marry}_{w'}\ \text{Afi}\big]$

b. Koku thinks λ_{w_x} Kofi wants $\lambda_{\underline{w'_{x'}}}$ $\big[\, [\text{LOG}_{\underline{x'}}\, [\boldsymbol{C}^{\boldsymbol{f'}}_{\underline{\boldsymbol{x'}}}\ (pro_i)]]\ \text{marry}_{w'}\ \text{Afi}\big]$

We point out a certain welcome consequence of this complex representations of the long-distance configuration in (66a). Pearson (2015: p.111, ex.91-92) shows that while long-distance-bound LOGP is read *de se* with respect to its antecedent the matrix attitude holder, it may be read in various *de re* ways with respect to the attitude of the intervening attitude holders (she demonstrates that using embedded double-vision scenarios; we leave out her example for space reasons). Let's see how this fact is precisely captured by the layered arrangement of counterpart operators in (66a).

The meaning the current system assigns to (66a) is approximated in (67). Due to LOG's presuppositional contribution, LOGP ends up referring to a *self*-counterpart of the matrix subject (Koku), guaranteeing the *de se* dependency with it.[25]

(67) *Simplified interpretation of (66a)*:
$\forall w_x \in \text{BEL}_{Koku}[\forall w'_{x'} \in \text{WANT}_{Kofi}[\text{the } f'\text{-counterpart of } x \text{ marries Afi in } w']]$[26]

The *de se* dependency, however, doesn't constrain how Kofi needs to be acquainted with Koku. Different ways to specify f' represent different ways for how Kofi is acquainted with Koku (according to Koku). If, for example, (Koku thinks that) Kofi is ordinarily acquainted with him through the description 'the person that I know as "Koku"', and it is this description which underlies f', then the result is the mundane construal in (68).

(68) *Possible construal of (67)*:
$\forall w_x \in \text{BEL}_{Koku}[\forall w'_{x'} \in \text{WANT}_{Kofi_{w_x}}$ [the person in w' that x' knows as "Koku" marries Afi in w']].
<u>Presupposition</u>: $\forall w_x \in \text{BEL}_{Koku}[x = \text{the person}_w \text{ Kofi knows as "Koku"}]$.[27]

But imagine a context where Kofi knows Koku only through the description 'my math teacher'—or at least, that this is what Koku believes to be the case. That is, Koku thinks

[24] If we didn't have the outermost counterpart in (65a), we would have the illegitimate configuration in which the subject of the most embedded predication resides in the matrix-bound worlds (w). Remember that given the ontology, an individual-denoting phrase needs to denote an individual that resides in the worlds where the local predication takes place, in this case w'.

[25] As mentioned, a more complete LF requires counterparts on the other individual-denoting phrases here. In particular different options for the resolution of Kofi's counterpart in the belief worlds of Koku (the matrix attitude holder) would yield different results per how Koku is acquianted with Kofi. This doesn't affect the point we are makin here in any substantial way.
We should note here that multiple embeddings on our account requires a certain modification of the definitions of 'Counterpart' and 'Source' from (35), to allow not just for actual individuals but also for possible individuals to have counterparts and to be Sources of others. The extension is straightforward but cannot be discussed for space reasons.

[27] The presuppositional statement comes about by the projection of the presupposition trigger LOG, which is embedded under two attitude predicates but is bound by the higher one.

"this guy Kofi doesn't know that my name is "Koku", he only knows me as his math teacher". This scenario specifies a different counterpart function than before, and the resulting construal is now:

(69) *Another possible construal of (67)*:
$\forall w_x \in \text{BEL}_{Koku}[\forall w'_{x'} \in \text{WANT}_{Kofi}$ [the person in w' that x' knows as his math teacher marries Afi in w']].
Presupposition: $\forall w_x \in \text{BEL}_{Koku}[x =$ the person$_w$ Kofi knows as his math teacher$]$.

The upshot is that in long-distance configurations, we predict that LOGP is restricted to refer *de se* to its antecedent (the matrix attitude holder), but can be read in various unconstrained ways with respect to the beliefs of intervening attitude holders—as is borne out by the discussion in Pearson (2015: 111).[28]

5.2 Competition between LOGP and the 1st person pronoun

We end with a proposal about the logophoric pronoun's restriction to occurrences in complements of attitude predicates. LOGPs cannot typically appear unembedded:

(70) * **yè**$_1$ dzó **Ewe**
LOGP leave
'I left. / He left.' (Pearson 2015: 78)

But on some assumptions detailed below, our approach currently predicts that (70) is grammatical and that the LOGP refers to the speaker, i.e. that (70) means "I left". LOGPs in the languages under discussion, however, cannot refer to the actual speaker; instead, there is a dedicated 1st person pronoun:

(71) **mè**$_1$ dzó **Ewe**
I leave
'I left.'

We offer a new perspective on this restriction, which crucially relies on our novel presuppositional semantics for LOGPs, together with the competition principle *Maximize Presupposition!* (Heim 1991 et seq.): in a nutshell, LOGP competes and loses to a 1ST-person pronoun whenever the latter can be used without change of meaning.

Previous proposals encode the restriction exemplified in (70) with the stipulation that LOGPs must be bound at the edge of complement clauses, by attitude verbs only (Heim 2005, von Stechow 2004, Pearson 2015), but we show below that our alternative account allows us to simplify the grammar of LOGPs somewhat and replace that stipulation with the weaker demand that LOGPs must be bound at the edge of some clause—not necessarily complement clauses.

We will then show that the *Maximize Presupposition!*-based proposal has an empirical advantage over the abovementioned accounts in that it can account for a hitherto unexplained generalization: even in attitude environments, LOGPs are out if anteceded by 1st-person attitude holders (Hagège 1974, Hyman and Comrie 1981).

[28] Although, Pearson (2015: 112) remarks in passing that LOGPs in such sentences only can, but don't have to, be construed *de se* when bound long-distantly. She does not provide the evidence, but that's in line with her general claim about LOGPs, see section 2.2.2). For us, *de se* with respect to the matrix holder is a necessity.

(72) a. *mè súsú bé **yè** dzó kábá. **Ewe**[30]
I think that **LOGP** leave early
'I think that I left early.'

b. mè súsú bé **mè** dzó kábá.
I think that **I** leave early
'I think that I left early.'

5.2.1 What LOGPs denote in matrix environments

Recall from section 4.3 the denotation we assign to a LOGP, repeated in (73).[31]

(73) a. $[\![\text{LOG}_x]\!] = [\lambda z : z = x.\ z]$

b. $[\![\text{LOGP}]\!]^g = [\![\text{LOG}_x\ pro_i]\!]^g = [\![pro_i]\!]^g$, defined only if $[\![pro_i]\!]^g = x$.

We assumed there (see fn. 13) that LOGPs—more accurately, the index on LOG—must be bound by an abstractor over centered-worlds from the edge of a clause.[32] If so, nothing prohibits a matrix occurrence of a LOGP where it is bound by the top-most centered-world abstractor introduced at the matrix level. The result would be:

(74) a. 'LOGP left' $\equiv\ \lambda_{w_{\underline{s}}} \left[[_{\text{LOGP}}\ \text{LOG}_{\underline{s}}\ pro_i]\ \text{left}_w\right]$

b. $[\![(74\text{a})]\!] = [\lambda_{w_{\underline{s}}} : [\![pro_i]\!]^g = \underline{s}.\ [\![pro_i]\!]^g \text{ left in } w]$

The world coordinate of a matrix centered-world corresponds, as standard, to a world compatible with the actual context of utterance, and it is natural to take the individual coordinate to be the speaker of the context. This is meant to capture the intuition that the 'logophoric center' of a matrix sentence is the speaker of the sentence. We technically implement this with the discourse rule in (75) relating the proposition expressed by a sentence to contexts in which the sentence can be appropriately uttered by speakers (the notions of Common Ground and Context Set in (75a) are adapted versions of their familiar predecessors from the works of Stalnaker (1978) et. seq.)

(75) a. A context is *appropriate* for a sentence ϕ if it determines a Context Set, a set of worlds c such that $c = \{w^* : w^*$ is compatible with the shared knowledge among the participants in the context$\}$.

b. A sentence ϕ uttered in context c by a speaker in c (notated '$sp(c)$') is true in a world $w* \in c$, iff: $[\![\phi]\!]^{g,c}(w^*_{sp(c)}) = 1$.

Given all this, (74a) is predicted to be grammatical and appropriately used to express the proposition that the speaker left, wrongly so (cf. (70)).

[30]Pearson (2015: p.97, fn.26) and Bimpeh (2023: :32, fn.7) both mention dialects of Ewe which allow LOGP to be anteceded by first-person pronouns, although the robustness of that data is not clear due to insufficient accessability to speakers of those dialects. We weren't able to verify that claim independently, but if correct, our account does not apply to such dialects.

[31]Counterpart operators are not needed in matrix environments, so they are left out here.

[32]If we didn't place any binding restrictions on LOG, we would wildly overgenerate readings, because then it wouldn't be clear what rules out a LOGP with a free-indexed LOG from referring to some contextually salient individual.

5.2.2 Competition via *Maximize Presupposition!*

In principle, we could avoid the problem with a syntactic stipulation to the effect that LOGPs are licensed only when bound by attitude predicates, like previous approches assumed (see section 3.1). But, as previewed, we want to offer an alternative that doesn't require this restriction, and to derive the infelicity of (70) as the result of competition with (71).

To do so we need a concrete analysis of 1st-person pronouns. We take the 1st-person pronoun to contain a variable pro_i and a semantic feature 1ST whose job is to restrict the variable's value to be the speaker, as represented in (76). This much is fairly standard (Heim 2008, Charnavel 2019, a.o.), but we also assume that 1ST, like LOG, comes with an index that must be bound by a logophoric abstractor. The index's value in this case is identified with pro_i and, unlike LOG, also with the actual speaker. The lexical entry of 1ST, a purely presuppositional function, is thus in (77a).[33] ('c' stands for the Context Set as was defined in (75a).)

(76) *LF of first-person pronoun*: $[_{\text{1STP}}\ \text{1ST}_s\ \text{pro}_i]$

(77) a. $[\![\text{1ST}_s]\!]^{g,c} = [\lambda z : \underline{z = s \wedge z = sp(c)}.\ z]$

b. $[\![\text{1STP}]\!]^{g,c} = [\![\text{1ST}_s]\!]^{g,c}([\![pro_i]\!]^{g,c}) =$
$[\![pro_i]\!]^{g,c}$, defined only if $[\![pro_i]\!]^{g,c} = s = sp(c)$.

Below is the LF and meaning of (71) 'I left'.

(78) 'I left.'

a. $\lambda_{w_s}\ \big[[_{\text{1STP}}\ \text{1ST}_s\ \text{pro}_i]\ \text{left}_w\big]$

b. $[\![(78\text{a})]\!]^{g,c} = [\lambda_{w_s} : \underline{[\![pro_i]\!]^g = s = sp(c)}.\ [\![pro_i]\!]^g \text{ left in } w.]$

The resulting partial proposition in (78b) has a stronger semantic presupposition—a strictly smaller domain—than the LOGP version of that sentence in (74b). While (74b) is in principle defined for any individual $s (= pro_i)$, (78b) is defined only for the actual speaker$(= pro_i)$. But when defined, both (74b) and (78b) have the exact same assertive content and therefore convey the same information: the speaker left. A situation where two alternative propositions have the same assertive content but one has a stronger presupposition is generally taken to feed the competition principle *Maximize Presupposition!* (Heim 1991, Percus 2006, Sauerland 2008, Schlenker 2012), with the consequence that the presuppositionally weaker alternative is blocked.

(79) *Maximize Presupposition!*.
Don't use a sentence ϕ in context S (cf. (75a)) if there is an alternative sentence ψ of ϕ such that the following are all met:

(i) ψ has stronger presuppositions than ϕ ($[\![\psi]\!]$'s domain is smaller than $[\![\phi]\!]$'s);

(ii) ψ's presuppositions are met in S (i.e. $w^*_{s^*} \in dom([\![\psi]\!]^S)$ for all $w^*_{s^*} \in S$);

(iii) ψ is contextually equivalent to ϕ ($\phi \equiv_S \psi$).

[33] This analysis of course is not applicable to languages with Indexical Shifting, as the 1st-person pronoun can also pick out attitude holders in those languages (see Deal 2020). In fact, if our analysis is to be extended to indexical shifting languages, then the first person pronoun in those languages can be analyzed the way we analyze LOGPs in West-African (and moreover indexical shifting languages would have to be assumed to lack a pronoun with the syntax and semantics in (76)-(77)).

We assume that 1STP is formally an alternative to LOGP (as it is syntactically no more complex than it; Katzir 2007) and therefore *Maximize Presupposition!* dictates that a sentence with LOGP cannot be used to refer to the speaker.[34] This explains the restriction in matrix environments.

5.2.3 1ST-LOG competition in embedded positions

1st-person pronouns are, of course, licensed not only in matrix positions but also in embedded ones. For e.g. *John thinks that I left*, a suitable analysis within our formalism is given in (80a), with its assigned interpretation in (80b). The presupposition of 1ST, once again, guarantees that the reference is to the (relevant counterpart of) the actual speaker. Note that the semantics of 1ST forces the index on 1ST to be bound by the matrix abstractor over centered-worlds, and that a counterpart operator must consequently be added.

(80) 'John thinks 1STP left'.

a. λ_{w_s} John thinks $\lambda_{w'_x}$ $\left[[_{\text{1STP}}\ C_x^f\,(\text{1ST}_s\ pro_i)]\ \text{left}_{w'}\right]$

b. $\lambda_{w_s}:\ \underline{[\![pro_i]\!]^{g,c} = s = sp(c)}.\ \forall w'_x \in \text{BEL}_{John}[C_x^f\,([\![pro_i]\!]^{g,c})\ \textit{left in } w']$

We can now proceed to suggest an explanation for the restriction on 1st-person antecedents, repeated from (72) (the facts hold in Yoruba and Igbo as well):

(81) LOGP*s cannot be anteceded by 1st-person attitude holders*:

a. *mè súsú bé **yè** dzó. **Ewe**
I think that **LOGP** leave
'I think that I left.'

b. mè súsú bé **mè** dzó.
I think that **I** leave
'I think that I left.'

Why is (81a) not acceptable? The answer, we propose, is that (81b) blocks (81a) by *Maximize Presupposition!*, just like (78b) blocked (74b) in matrix environments. Here is the LF-analysis of the two sentences:

(82) 'I think LOGP left'.

a. λ_{w_s} I think $\lambda_{w'_x}$ $\left[[_{\text{LOGP}}\ C_x^f\,(\text{LOG}_s\ pro_i)]\ \text{left}_{w'}\right]$

b. $\lambda_{w_s}:\ \underline{[\![pro_i]\!]^{g,c} = s}.\ \forall w'_x \in \text{BEL}_{sp(c)}[C_x^f\,([\![pro_i]\!]^{g,c})\ \textit{left in } w']$

[34] *Maximize Presupposition!* has been invoked to explain certain inferences arising from the use of the presuppositionally weaker alternative ('Anti-presupopositions'; Percus 2006 a.o.). In our case, however, no Anti-presupposition can be detected from a use of unembedded LOGP because the relevant presupposition of the stronger alternative 1STP is always met; more precisely, a hypothetical context in which that presupposition isn't met would be one where the sentence as parsed in (78a) could not be uttered by the speaker in the context to begin with, given the definition of appropriate context in (75).

(83) 'I think 1STP left'.

a. λ_{w_s} I think $\lambda_{w'_x}$ $\big[[_{1\text{STP}}\ C^f_x\,(1\text{ST}_s\ pro_i)]\ \text{left}_{w'}\big]$

b. $\lambda_{w_s}:\ \underline{[\![pro_i]\!]^{g,c} = s = sp(c)}.\ \forall w'_x \in \text{BEL}_{sp(c)}[C^f_x\,([\![pro_i]\!]^{g,c})\ \textit{left in } w']$

The underlined parts reveal again that (83) has a strictly stronger presupposition than (82). But they have the exact same assertive content. Therefore the latter is blocked by *Maximize Presupposition!*.[35]

While this competition-based account explains (81) as a direct extension of the account of matrix environments, it is difficult to see how previous accounts of the dependency between a LOGP and its antecedent can capture it without further stipulations. If LOGPs are merely syntactically required to be bound by an attitude verb (as is assumed by Simplex Binding, section 3.1), then additional mechanisms are needed to ensure that a 1st-person subject is not a possible antecedent.[36]

6 Conclusion

This paper provided evidence that logophoric pronouns (LOGPs) in Ewe, Yoruba and Igbo support both strict and sloppy readings in ellipsis configurations and in sentences with *only* (following observations in Culy 1994 and Bimpeh and Sode 2021), and offered a formal analysis that could capture this behavior. The account supplants existing accounts of LOGPs with the idea that LOGPs are pronouns that contain a semantic feature LOG in charge of encoding the *de se*-reference to the attitude holder (following Bimpeh et al. (2024)), but whose contribution can be ignored at the level of focus alternatives, like other pronominal features. We also showed how long-distance dependencies of LOGPs are handled, and provided a novel solution for the restriction on the occurrence of LOGP in matrix environments and with 1st-person antecedents.

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[35]In (82a) the index on LOG is bound by the matrix abstractor (λ_{w_s}). Things change if we choose the index x bound by the embedded abstractor ($\lambda_{w'_x}$). That parse, in (84), generates the presupposition underlined in (84b).

(84) a. λ_{w_s} I think $\lambda_{w'_x}$ $\big[[_{\text{LOGP}}\ \text{LOG}_x\,[C^f_x\ pro_i]]\ \text{left}_{w'}\big]$

b. $\lambda_{w_s}:\ \underline{\forall w_x \in \text{BEL}_{sp(c)}, [\![C^f_x\ pro_i]\!]^g = x}.\ \forall w'_x \in \text{BEL}_{sp(c)}[C^f_x\,([\![pro_i]\!]^{g,c})\ \textit{left in } w']$

This amounts to the presupposition that across the speaker's belief worlds, her *self*-counterpart is also the counterpart of *pro*. We are not sure at the moment what blocks this representation.

[36]The logic of the account of (81) leads us to expect that LOGPs are fine with 2nd-person antecedents. While this is correct for Ewe (Clements 1975, Bimpeh 2023) and other languages (see Hyman and Comrie 1981, Stirling 1993, Nikitina 2012), it appears to not be true for all logophoric languages (see e.g., Kiemtoré 2022). We leave this issue unresolved for now and hope to revisit it in a future occasion.

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A Appendix

A.1 Language profiles

All three languages are spoken in West Africa and belong to the Niger-Congo language families. Ewe (indigenously pronounced Eʋe) is a Niger-Congo language of the Kwa group that is a member of the larger unit of closely related languages known as the Gbe languages. According to Capo (2010) and Ameka (1991), Gbe 'voice / language' refers to a dialect cluster comprising Gen, Aja, Xwla-xweda (Phla-Pherá), Fon, and Ewe. Speakers of the Gbe languages inhabit the Volta and Oti regions of Ghana, the southern part of Togo, the southern part of Benin, as well as parts of Ogun and Lagos state, Nigeria. Ewe is a cover term used to refer to a group of (sub)dialects spoken in the Volta and Oti regions of Ghana by about 2.5 million people, and in the southern part of Togo by approximately 1 million people.

Yoruba is a dialect continuum spoken in the area that covers Western Nigeria and Lagos, the Ilorin province of Northern Nigeria, parts of Benin, Sierra Leone, Cote D'Ivoire and the Gambia. The language belongs to the Benue-Congo language family and is spoken by 44.6 million speakers (Eberhard et al. 2020). According to Bamgboṣe (1966), Yoruba comprises about twenty dialects such as Ijèbú, Egbá, Ijèsà, Oyó, Owó, Ondó, each of which differs considerably from the other phonologically, and lexically. There is a standard variety of Yoruba, based on the Oyó dialect, called Koine which is used for the purpose of education, writing, and contact between persons of different dialects.

Igbo is the principal native language of the Igbo people, an ethnic group from Southeastern Nigeria, and the second most populous indigenous language of that area. Other Igbo speaking areas include Equitorial Guinea and Cameroon. The Igbos are one of the three largest ethnic groups in Nigeria (Awde and Wambu 1999). They are predominately located in five states namely, Anambra, Imo, Ebonyi, Abia, and Enugu. A significant number of Igbo communities can also be found in neighbouring states such as Rivers, Asaba as well as the commercial centre of the country, i.e., Lagos (Awde and Wambu 1999). The language belongs to the Benue-Congo language family and is spoken by 2.2 million speakers (Eberhard et al. 2020). There are dozens of geographical dialects. These dialects differ in certain grammatical, lexical and phonological details (Emenanjo 2015). The standard dialect is based mainly on Owerri, Umuahia and Onitsha dialects.

A.2 Obligatory *de se* readings of logophors

This section provides more details and data on the elicitation of *de re* vs. *de se* scenarios. As discussed in section 2 of the main paper, one aspect of our methodology includes the inclusion of ORDPs in our target sentences when tested in *de re* contexts (mistaken identity scenarios).

Another modification on our side is the fact that both LOGPs and ORDPs were tested in *de se* scenarios. The *de se* contexts were designed to be minimally different from the *de re* contexts. With *think* and *say* specifically, we created contexts which were equally elaborate in order to test whether the complexity of the scenario influences the judgement. The complexity did not interact with the judgements of our speakers. We present the results for the *de se* contexts in (85)-(88), along with the *de re* counterparts, repeated from above, in (89)-(92).[37] Interestingly, languages differ regarding the availability of ORDP in *de se* contexts. Whereas in Ewe ORDP is unacceptable and only the logophor is licensed, in Yoruba and Igbo both LOGP and ORDP are acceptable. Given the data in (85)-(88), in particular the availability of ORDP in Yoruba and Igbo, we conclude that pronouns can indeed be ambiguous between *de se* and *de re* readings, and if they are, our methodology will track such ambiguity. In other words, we can exclude the possibility that logophors might be ambiguous after all, but a general bias for pronouns towards *de se* readings makes them unacceptable in *de re* contexts (cf. Pearson 2022). If this were the case, we would not expect ORDP to be acceptable in both *de se* and *de re* contexts in Yoruba and Igbo.[38]

(85) <u>*De se context, think:*</u> *(Bimpeh et al. 2024)*
Donald Duck (DD) went to the grocery store to buy flour. Then, he mistakenly put sugar in his cart. DD went on and then, he saw a trail of sugar going up and down the aisles and thought that someone's bag had a hole in it and looked around for the guy. DD says: "I wonder who is losing sugar" "Certainly, the guy who is losing sugar is stupid, and it is not me because I bought flour not sugar!" Later he says "But I did not check!" "Let me see if it's me the stupid guy who is losing sugar." He checks in his bag and sees the sugar. Finally, he realised.

a. Donald Duck súsú be **yè** / **#é** dzɔ-mo-vi. **Ewe**
Donald Duck think that LOGP / ORDP exist.with-face-small
'Donald Duck thinks that he is stupid.'

b. Donald Duck chèrè nà **yá** / **ọ́** bụ̀ ónyéńzúzù. **Igbo**
Donald Duck think that LOGP / ORDP COP stupid.person
'Donald Duck thinks that he is stupid.'

c. Donald Duck rò pé **òún** / **ó** jẹ́ òmùgọ̀. **Yoruba**
Donald Duck think that LOGP / ORDP COP stupid.person
'Donald Duck thinks that he is stupid.'

(86) <u>*De se context, say:*</u>
Elmo goes to visit Big Bird. While there, Big Bird shows him old paintings he found from back when Elmo was living there with him. After looking at several pictures, Elmo does not recognize one of the paintings which is particularly pretty. Elmo says: "I wonder who painted this. Certainly, the person who painted was a good painter and it is not me, because I'm not very talented in painting!" Later he says "But I did not check! Let me see if I'm the good painter who painted this" Looking carefully at the painting he sees his initials and he says "Oh look there is my signature on it!". Finally he realized.

[37] Note that we are extending our data set to the matrix verb *promise* in (88) and (92) with additional data points which confirm our hypothesis. These data points are not included in the main paper since we did not investigate strict and sloppy readings of logophors under *promise* as consistently as with the other predicates.

[38] For a way to analyze the cross-linguistic split in *de se* scenarios, see Bimpeh et al. (2024).

a. Elmo bé **yè** / #**é** nyé nùtálá nyùìé a dé. **Ewe**
Elmo say LOGP / ORDP is thing.draw.one-who good INDF
'Elmo said that he is a good painter.'

b. Elmo sị nà **yá** / **ọ́** nà-ésè íhé ǹké ọ́má. **Igbo**
Elmo say that LOGP / ORDP IPFV-paint thing nke good
'Elmo said that he is a good painter.'

c. Elmo so pé **òún** / **ó** jẹ́ akunlé tí ó dára. **Yoruba**
Elmo say that LOGP / ORDP is painter REL RP good
'Elmo said that he is a good painter.'

(87) <u>*De se context, hope:*</u>
Goofy is a candidate in the election. He wants to get elected.

a. Goofy lè mɔ́-kpɔ́-m bé **yè** / #**é** à ɖù àkɔ̀-dádá lá dzí.
Goofy COP path-see-PROG that LOGP / ORDP IRR win vote-cast DEF post
'Goofy hopes that he will win the election.' **Ewe**

b. Goofy nà èlé ánya nà **yá** / **ọ́** gà è-mérí n' àtúḿvóòtù áhù.
Goofy have look eye that LOGP / ORDP FUT PTCP-win in election DEM
'Goofy hopes that he will win the election.' **Igbo**

c. Goofy ń rètí pé **òun** / **ó** yoo borí nínú ìbò náà.
Goofy PROG hope that LOGP / ORDP FUT win in election DET
'Goofy hopes that he will win the election.' **Yoruba**

(88) <u>*De se context, promise:*</u>
Kofi/Aki/Adé broke his mom's very expensive grinding bowl. He was very sorry. He decided to make it up to her. He made a promise to his mom that he would buy a new one.

a. Kofi doŋugbe be **yè** / #**é** a jle ʋe gba yéyê.
Kofi promise that LOGP / ORDP IRR buy earthen ware new
'Kofi promised that he will buy a new grinding bowl.' **Ewe**

b. Ákí kwèrè ńkwà nà **yá** / **ọ́** gà à-zù éféré ògwùgwé óhùru.
Aki promise promise that LOGP / ORDP FUT PTCP-buy bowl grinding new
'Aki promised that he will buy a new grinding bowl.' **Igbo**

c. Adé se ìlérí wípé **òun** / **ó** maa ra abọ́ ọlọ tuntun.
Ade make promise that LOGP / ORDP FUT buy bowl grinder new
'Adé promised that he will buy a new grinding bowl.' **Yoruba**

(89) <u>*De re context, think:*</u> *(Bimpeh et al. 2024)*
Donald Duck (DD) went to the grocery store to buy flour. He mistakenly put sugar in his cart. DD then saw a trail of sugar going up and down the aisles and thought that someone's bag had a hole in it and looked around for the guy. DD says: "I wonder who is losing sugar; certainly, the guy who is losing sugar is stupid, as he does not check his

bag". Later he says: "Is it me the stupid guy who is losing sugar?" "No, because I did not buy sugar but flour".

a. Donald Duck súsú bé **#yè** / **é** dzɔ̀mɔ̀ví. **Ewe**
Donald Duck think that LogP / OrdP stupid.person
'Donald Duck thinks that he is stupid.'

b. Donald Duck chèrè nà **#yá** / **ọ́** bụ̀ ónyéńzúzù. **Igbo**
Donald Duck think that LogP / OrdP COP stupid.person
'Donald Duck thinks that he is stupid.'

c. Donald Duck rò pé **#òun** / **ó** jẹ́ òmùgọ̀. **Yoruba**
Donald Duck think that LogP / OrdP COP stupid.person
'Donald Duck thinks that he is stupid.'

(90) *De re context, say:* *(Bimpeh et al. 2024)*
Elmo goes to visit Big Bird. While there, Big Bird shows him old paintings he found from back when Elmo was living there with him. After looking at several pictures, Elmo does not recognize one of the paintings which is particularly pretty. Elmo says: "I wonder who painted this. Certainly, the person who painted is a good painter". Later he says: "Is it me the good painter who painted this? No, because I am not very talented in painting".

a. Elmo bé **#yè** / **é** nyé nùtálá nyùìé a dé. **Ewe**
Elmo say LogP / OrdP is thing.draw.one-who good INDF
'Elmo said that he is a good painter.'

b. Elmo sị nà **#yá** / **ọ́** nà-ésè íhé ǹké ọmá. **Igbo**
Elmo say that LogP / OrdP IPFV-paint thing nke good
'Elmo said that he is a good painter.'

c. Elmo so pé **#òún** / **ó** jẹ́ akunlé tí ó dára. **Yoruba**
Elmo say that LogP / OrdP is painter REL RP good
'Elmo said that he is a good painter.'

(91) *De re context, hope:*
Goofy is so drunk that he has forgotten he is a candidate in the election. He watches someone on TV and finds that that person is a terrific candidate, who should definitely be elected. Unbeknownst to Goofy, the candidate he is watching on TV is Goofy himself.

a. Goofy lè mɔ́-kpɔ́-m bé **#yè** / **é** à ɖù àkɔ̀-dádá lá dzí.
Goofy COP path-see-PROG that LogP / OrdP IRR win vote-cast DEF post
'Goofy hopes that he will win the election.' **Ewe**

b. Goofy nà èlé ánya nà **#yá** / **ọ́** gà è-mérí n' àtúḿvóòtù áhù.
Goofy have look eye that LogP / OrdP FUT PTCP-win in election DEM
'Goofy hopes that he will win the election.' **Igbo**

c. Goofy ń rètí pé **#òun** / **ó** yoo borí nínú ìbò náà.
Goofy PROG hope that LogP / OrdP FUT win in election DET
'Goofy hopes that he will win the election.' **Yoruba**

(92) *De re context, promise:*
Kofi/Aki/Adé threw a party with friends last night at his parents' house. He got so drunk that he broke his mom's expensive grinding bowl. In the afternoon when he woke up, he saw the broken grinding bowl to his horror but totally forgot it was him who broke it. He thought it was someone else. The next day his mom comes home, sees the mess with the grinding bowl and starts yelling at him. Kofi/Aki/Adé tries to calm her down with these words: "I promise you that whoever broke your grinding bowl will pay for a new one."

a. Kofi doŋugbe be **#yè** / **é** a jle ʋe gba yéyê.
Kofi promise that LogP / OrdP IRR buy earthen ware new
'Kofi promised that he will buy a new grinding bowl.' **Ewe**

b. Ákí kwè-rè ńkwà nà **#yá** / **ọ́** gà à-zù éféré ògwùgwé óhùru.
Aki promise-PST promise that LogP / OrdP FUT PTCP-buy bowl grinding new
'Aki promised that he will buy a new grinding bowl.' **Igbo**

c. Adé se ìlérí wípé **#òun** / **ó** maa ra abọ́ ọlọ tuntun.
Ade make promise that LogP / OrdP FUT buy bowl grinder new
'Adé promised that he will buy a new grinding bowl.' **Yoruba**

We further investigated whether the *de re* reading becomes available if LogP is focused, following a suggestion in Pearson (2017, 2022). This was based on the observation for English 'he himself' which is often argued to only receive a *de se* reading, though when focused can be read *de re*, see (93).

(93) *Context: Two speakers, A and B. A says:* (Pearson 2017)
John sees himself in a mirror, but fails to recognize that it's him. He thinks, 'that guy is an idiot', but not 'I am an idiot'.
B says: I don't understand the story. Who does John think is an idiot?
A says: John thinks that he himself is an idiot.

In (94), we present a modified version of the *de re* context (6) and the results for the target sentences in Ewe. We can report that the adjusted scenario did not make the logophor more available. Indeed, the judgements in (94a) and (94b) do not differ from the judgements of the target sentences in (6). Since in Ewe focus is marked morphologically, we also investigated the acceptability of an overtly marked focused logophor in (94c). This sentence too was judged infelicitous.

(94) *Context: Two speakers, A and B. A says:*
Donald Duck (DD) went to the grocery store to buy flour. He mistakenly put sugar in his cart. DD then saw a trail of sugar going up and down the aisles and thought that someone's bag had a hole in it and looked around for the guy. DD says: "I wonder who is losing sugar; certainly, the guy who is losing sugar is stupid, as he does not check his bag". Later he says: "Is it me the stupid guy who is losing sugar?" "No, because I did not buy sugar but flour".
B says:
I don't understand what happened. Who does Donald Duck think is stupid?
A says:

a. #Donald Duck súsú bé **yè** dzɔ̀mòví. **Ewe**
Donald Duck think that LOGP stupid.person
'Donald Duck thinks that he is stupid.'

b. Donald Duck súsú bé **é** dzɔ̀mòví.
Donald Duck think that ORDP stupid.person
'Donald Duck thinks that he is stupid.'

c. #Donald Duck súsú bé **yè-yé** dzɔ̀mòví.
Donald Duck think that LOGP-FOC stupid.person
'Donald Duck thinks that it is he who is stupid.'

We conclude that at least for Ewe focus does not facilitate a *de re* reading of logophors. As for Yoruba and Igbo, testing scenarios like (93) is confounded by the fact that the logophor is form-identical to emphatic pronouns in these languages. In (95), we can observe that the forms *òun* and *yá* can occur in unembedded environments if they are focused, signaled in (95a) with the focus marker in Yoruba, and in (95b) by the presence of a cleft sentence in Igbo.

(95) a. **Òun** ni Tọ́lá rí. **Yoruba**
3SG.EMPH FOC Tola see
'It is her/him/it that Tola saw.' (Manfredi 1987: 103)

b. Ọ́ wụ́ **ya** kà ḿ hụ̀-rụ. **Igbo**
EXPL be 3SG.EMPH that 1SG see-ASP
'It is her/him/it that I saw.' (Manfredi 1987: 107-108)

Since the context in (93) is designed to trigger the occurrence of a focused pronoun (due to the added question by speaker B) and focused pronouns are identical in form to LOGP in Yoruba and Igbo, the results of the adjusted scenario would not be very informative. In other words, we would not be able to distinguish focused LOGP from focused ORDP by purely looking at the morphological form since in these dedicated positions the forms are neutralized.

While the fact that some languages display an interaction of focus and logophoricity certainly warrants further investigation, we believe these observations are orthogonal to the availability of *de re* readings of logophors in focus. Rather, what we should ask is why in some languages the same morphological form realizes LOGP and focused ORDP.[39]

A.3 Ewe *yè* vs. *yè-a*

Satık (2019, 2020, 2021) identifies *yè-a* in Ewe as the non-finite counterpart of *yè*, where the former is a form that appears under what we cross-linguistically categorize as Obligatory Control (OC) predicates and is essentially *yè* undergoing coalescence with the irrealis marker. The claim is that *yè-a* constitutes overtly spelled out PRO (where the irrealis marker signals non-finiteness), whereas *yè* is a logophor that can appear in finite clauses. Satık provides data in support of this claim wrt. *de re* readings and long distance antecedents: like PRO *yè-a* cannot take long distance antecedents and only allows for *de se* readings, whereas *yè* can take long distance antecedents and can be read *de se* and *de re*. As is evident throughout our paper, our results question Satık's observations and proposal. For example, we provide two mistaken identity scenarios, one for the attitude verb 'hope' requiring *yè-a* in Ewe and one for 'think', which

[39] One language where this interaction becomes remarkably transparent is Jula, a Mande language spoken in West Africa, see Kiemtoré (2022) for discussion.

tends to occur with embedded *yè*. Both forms, however, only receive *de se* readings for our consultants. Similarly, in the multiple embedding structures both *yè-a* and *yè* can take long distance antecedents. So at least for our speakers, the irrealis marker does not seem to indicate whether *yè* is a logophor or spells out OC PRO.

Given that the type of attitude predicate consistently varies in Satık's data, in the sense that embedded clauses with *yè* occur with predicates typically selecting finite clauses, and embedded clauses with *yè-a* occur with predicates typically selecting non-finite clauses (at least in English), it is not easy to tell if the behaviour of *yè-a* is different from *yè* due to the irrealis marker or due to the embedding predicate (or both). Fortunately, however, we could observe with our speakers that the irrealis marker *á* can indeed optionally occur under 'say' and 'think', and when it does it signals future orientation. In (96), we show the effect of the irrealis marker and signal the interpretive effect in the translation.

(96) *The irrealis marker in **Ewe***

a. Kòkú$_1$ bé **yè**$_{1/*2}$ lɔ̃ Àfí.
Koku say LOG love Afi
'Koku said that he loves Afi.'

b. Kòkú$_1$ bé **yè**$_{1/*2}$ **á** lɔ̃ Àfí.
Koku say LOG IRR love Afi
'Koku said that he will love Afi (one day).'

c. Kòkú$_1$ súsú bé **yè**$_{1/*2}$ lɔ̃ Àfí.
Koku think that LOG love Afi
'Koku thinks that he loves Afi.'

d. Kòkú$_1$ súsú bé **yè**$_{1/*2}$ **á** lɔ̃ Àfí.
Koku think that LOG IRR love Afi
'Koku thinks that he will love Afi (one day).'

Since the contrasts in (96) provide better minimal pairs to investigate the contribution of the irrealis marker isolated from the embedding predicate, we can conduct a small investigation based on the verbs 'say' and 'think'. First, we test long distance antecedents. As is shown in (97), neither combination of 'say' and 'think' leads to an observation where *yè-a* only takes the local attitude holder as an antecedent. If we compare the data in (97) with the multiple embedding structures in the main paper, we find that the addition of the irrealis marker does not restrict *yè* to co-refer only with the local antecedent.

(97) *Long distance antecedents with yè-a in **Ewe***

a. Kòkú$_1$ súsú bé Kòfí$_2$ bé **yè**$_{1/2}$ **á** dè Àfí.
Koku think that Kofi say LOG IRR marry Afi
'Koku thinks that Kofi said that he will marry Afi.'

b. Kòkú$_1$ bé Kòfí$_2$ súsú bé **yè**$_{1/2}$ **á** dè Àfí.
Koku say Kofi think that LOG IRR marry Afi
'Koku says that Kofi thinks that he will marry Afi.'

Let us now turn to *de re* readings. First, we have to modify the mistaken identity context for the attitude verb 'say' in (7) with a future orientation to make the occurrence of the irrealis marker felicitous, which is given in (98). As can be seen by the judgements of the target sentences, no effect of the irrealis marker was detected. Importantly, the judgements for LOGP and ORDP are the same in (7) and (98).

(98) <u>*De re context, say, future orientation:*</u>
Big Bird teaches painting to his friends. In the past, Elmo took one of Big Bird's classes and he painted some paintings. One day, Elmo goes to visit Big Bird. While there, Big Bird shows him one painting that he found in the house, which is particularly pretty. Elmo does not recognize that this is one of the paintings he made when he took Big Bird's class. So he says: "This painting is wonderful. Whoever painted this has a great talent. This person will be a good painter one day."

a. #Elmo	bé	**yè**	**á**	zù	nùtálá	gã	gbè	áɖé	gbè.
Elmo	say	LOG	IRR	become	thing.draw.onewho	big	day	INDEF	day

'Elmo says that he will be a great artist one day.'
Comment: Not good because Elmo is speaking about himself.

b. Elmo	bé	**é**	**á**	zù	nùtálá	gã	gbè	áɖé	gbè.
Elmo	say	3SG	IRR	become	thing.draw.onewho	big	day	INDEF	day

'Elmo says that he will be a great artist one day.'
Comment: This is good because Elmo is refering to whoever did the painting even though we all know that Elmo is the one who painted it.

We conclude from our small investigation that at least for our Ewe speakers, the irrealis marker by itself does not indicate a non-finite clause, in which *yè* spells out OC PRO, as is proposed in Satık (2020, 2021).[40]

If it is not the irrealis marker that determines the status of what is spelled out by *yè*, what else could cause the different judgements in Satık's data? It is, after all, possible that the differences stem from the fact that Satık's examples often make use of attitude predicates which are different from ours. For example, the mistaken identity scenario Satık tested for *yè-a* included the Ewe verb for 'expect' (Satık 2021: 7), and the long distance antecedent data were based on the Ewe verbs for 'try' and 'want' (Satık 2021: 8). The predicates tested in our study are 'say', 'think', 'hope', 'promise', and 'want'. While we have shown that the irrealis marker does not indicate an OC profile per se, it is still possible that certain predicates impose additional OC restrictions, which could for example influence the ability of the embedded subject to take a long distance antecedent. While we did not find this restriction with our speakers for 'want', such a restriction under 'want' is documented in Gengbe, a Niger-Congo language closely related to Ewe and spoken in Southern Togo and Benin (Grano and Lotven 2019: p.484, ex.8). Hence, we do not rule out dialectal differences with certain attitude verbs.

A.4 Logophors vs. exempt anaphors

The notion of logophoricity has played an important role in analyses of so-called exempt anaphora (Sells 1987, Reinhart and Reuland 1993, Charnavel 2020a,b). In this section, we will briefly discuss to what extent logophoric pronouns in West-African languages and exempt

[40]The connection between the irrealis marker and non-finiteness is in some sense also critically discussed in Satık's own work (Satık 2021: 5-6). First, non-finiteness is not so easy to investigate given that Ewe does not display subject-verb agreement or tense marking. Second, the irrealis marker can also appear in matrix clauses.

anaphors can be captured with the same analysis.[41] Our preliminary conclusion is that they should not receive the same analysis.

First let us consider some properties that exempt anaphora share with logophoric pronouns. For one, strict readings are attested for exempt anaphors (see Lebeaux 1984, Reinhart and Reuland 1993, Charnavel 2020a). This is shown with VP-ellipsis and English *himself* acting as an exempt anaphor in (99).

(99) John_1 thought there were some pictures of $\textbf{himself}_1$ inside, and Bill did, too.

$\rightsquigarrow_{sloppy}$ Bill thought that there were some pictures of himself inside too.

$\rightsquigarrow_{strict}$ Bill thought that there were some pictures of John inside too.

(Lebeaux 1984: 346)

Moreover, exempt anaphors can only receive *de se* readings, as is exemplified with the Italian exempt anaphor *proprio* in (100) and (101), taken from Chierchia (1989). The exempt anaphor is licensed in a *de se* context in (100), but is infelicitous in the mistaken identity context set up in (101).

(100) *Context de se:* Pavarotti sees a man in the mirror with his pants on fire. After a while he realizes for obvious reasons that the man in the mirror is he himself. Pavarotti thinks: "My pants are on fire."

Pavarotti	crede	che	i	**propri**	/	suoi	pantaloni	siano	in	fiamme.
Pavarotti	believes	that	DEF	POSS.SELF	/	POSS	pants	are	on	fire

Pavarotti believes that his pants are on fire.

(101) *Context non de se:* Pavarotti sees a man in the mirror with his pants on fire. The man happens to be Pavarotti himself but he doesn't realize this. Pavarotti thinks: "His pants are on fire."

Pavarotti	crede	che	i	#**propri**	/	suoi	pantaloni	siano	in	fiamme.
Pavarotti	believes	that	DEF	POSS.SELF	/	POSS	pants	are	on	fire

Pavarotti believes that his pants are on fire.

Charnavel (2020a,b) proposes that exempt anaphors should be considered a subcase of local anaphors, thereby capitalizing on the observation that they show identical morphological realization. The exempt status is derived by the assumption that anaphors can be bound by a silent covert pronoun which is introduced by a silent logophoric operator, shown in (102).[42] This pronoun is co-indexed with the antecedent, and thus can trigger strict or sloppy readings. Based on Charnavel and Sportiche (2016), Charnavel develops an analysis in which both pronoun and operator form a clausal projection (notated here LGP) which is merged at the periphery of the binding domain for Condition A (*YP* in (102a)). The operator in (102b) ensures obligatory *de se* readings.

[41] For the data in this section, we consulted one speaker per language.

[42] We discuss the account by Charnavel (2020a,b) here specifically, but note that there are many versions of a silent operator account which have been proposed for logophors in particular (Koopman and Sportiche 1989, Speas 2004, Safir 2004, Adésolá 2005), but also PRO (Chierchia 1989), exempt anaphors and logophors (Anand 2006, Baker and Ikawa 2024), indexiphors (Deal 2018) and perspectival anaphora (Sundaresan 2018).

(102) Charnavel (2020b: 9,36)

a. ... DP_i ... $[_{YP}$ $[_{\text{LGP}}$ pro_{log-i} OP_{log} $[_{\alpha}$ ... ANAPHOR$_i$ ...]]]

b. $[\![OP_{log}]\!] = \lambda\alpha\lambda x.\alpha$ from x's perspective

While an analysis of logophoric pronouns in Ewe, Yoruba and Igbo in terms of exempt anaphors along the lines of Charnavel's (2020a, 2020b) theory cannot a-priori be ruled out, we think that there are two central problems with it. The first comes from the morphology of logophors. The elegance of Charnavel's theory stems from the fact that exempt anaphors are subsumed under standard local anaphora in languages like English, French, Japanese etc. This is desirable as they share the morphology with local anaphora in these languages. However, at least in Ewe and Yoruba, the reflexive pronoun looks radically different from the logophoric pronoun, shown in (103) and (104).[43] Hence, it is not clear what would motivate a unification of the grammar of LOGPs with that of a exempt anaphors in these languages.

(103) *Reflexives in Ewe* (Clements 1975)

a. Kòkú kpɔ́ **é-dòkùì** / ***yè**.
Koku saw **3SG-REFL** / **LOGP**
'Koku saw himself.'

b. Kòkú bé yè kpɔ́ yè-dókùì.
Koku say LOG see LOG-REFL
'Koku said that he saw himself.'

(104) *Reflexives in Yoruba* (Lawal 2006)

a. Taiwo féron **araare**.
Taiwo like **REFL**
'Taiwo likes himself.'

b. *Taiwo féron **óun**.
Taiwo like **LOGP**
'Taiwo likes himself.'

The second problem we see in unifying logophors with exempt anaphors is based on the syntactic and semantic distribution. The licensing conditions of LOGPs appear to be different in complex ways from those of exempt anaphors. Specifically, LOGPs seem to be more narrowly licensed. For example, it is known that an exempt anaphor in English need not necessarily be c-commanded by a suitable antecedent (Ross 1970, Lebeaux 1984, Charnavel 2020a: chapter 1.1), as is shown in (105).

(105) *Exempt anaphor in English*
Mary's anger sometimes turns against **herself**.

[43] In Igbo, the form of the logophor *yá* is identical to the morphological form of the pronoun in object position. Reflexives like 'herself' are formed by *ònwé* + *yá* (Uchechukwu 2011). While this could signal that the reflexive and the logophor share a morpheme, it is more likely that the *yá* in *ònwé yá* is simply the pronoun marked for accusative, similar to English *her-self*.

In the same position, however, the logophoric pronoun in Ewe is unacceptable (106a). Instead, either the strong 3rd person pronoun (106b) or a reflexive (106c) is used.[44]

(106) *Probing for exempt anaphor in Ewe*

a. *Kòfí ʄé dzìkú lè ɣèɣíɣí-áɗé-wó-mè trɔ́-ná ɗé **yè** ŋùtɔ́ dzi.
Kofi POSS anger be times-INDEF-PL-inside turn-HAB onto **LOGP** body top
'Kofi's anger sometimes turns against himself.'

b. Kòfí ʄé dzìkú lè ɣèɣíɣí-áɗé-wó-mè trɔ́-ná ɗé **éyà** ŋùtɔ́ dzí.
Kofi POSS anger be times-INDEF-PL-inside turn-HAB onto **3SG** body top
'Kofi's anger sometimes turns against himself.'

c. Kòfí ʄé dzìkú lè ɣèɣíɣí-áɗé-wó-mè trɔ́-ná ɗé **édókùì** dzi.
Kofi POSS anger be times-INDEF-PL-inside turn-HAB onto **REFL** top
'Kofi's anger sometimes turns against himself.'

A similar observation can be made for LOGP in Yoruba. In contrast to the English exempt anaphor, the logophoric pronoun is unacceptable in the same position (107a). Instead the pronoun is used (which in this example is assimilated to the preceding vowel), see (107b).

(107) *Probing for exempt anaphor in Yoruba*

a. *ìbínú u Kọ́lá má kóbá **óun** nígbàmíràm
anger POSS Kola do affect **LOGP** sometimes
'Kola's anger sometimes turns against himself.'

b. ìbínú u Kọ́lá má kóbá **a** nígbàmíràm
anger POSS Kola do affect **3SG** sometimes
'Kola's anger sometimes turns against himself.'

The distributional differences align with a recent study by Baker and Ikawa (2024), henceforth B&I, in which a comparison is drawn between a West-African type logophor (Ibibio *ímọ̀*) and an exempt anaphor (Japanese *zibun*) across various environments. They encounter several differences between the two types, one is related to distribution. Concretely, they observe that Japanese *zibun*, but not Ibibio *ímọ̀*, is licensed in a root clause if an adjunct like 'according to X' or 'in X's opinion' is present, see (108).

(108) *Logophor vs. exempt anaphor* (Baker and Ikawa 2024: 900)

a. *Ke akikere Okon, **ímọ̀** i-ma i-due. *Ibibio*
in thought Okon **LOGP** 3.LOG-PST-3.LOG-guilty
'In Okon$_i$'s opinion, Emem/he$_i$ was guilty.'

b. Taroo-ni.yoruto **zibun**-wa waruku-nai-?(n(o)-da-)soo-da. *Japanese*
Taroo-according.to **self**-TOP bad-NEG-no-COP-EVID-COP
'According to Taroo$_i$, self$_i$ is not bad.'

[44]Note that Ewe seems to display both LOGP and and what can presumably be classified as an exempt anaphor in (106c), which strengthens the suspicion that they are not exactly the same creature.

The Ewe logophor *yè* patterns with the logophor in Ibibio and not with the exempt anaphor in Japanese, as the following examples show.[45]

(110) *B&I context (108) in Ewe*

a. Lè Òkón$_1$ ƒé sùsú mè lá, **Kòfí** / **é**$_{1/2}$ / ***yè** xɔ̀sè bé yè nyá àgbàlè.
in Okon POSS mind inside TOP **Kofi** / **3SG** / **LOGP** believe that LOG know book
'In Okon's mind, Kofi/he believes he is smart.'

b. Lè Òkón$_1$ ƒé sùsú mè lá, **éyà yé** / ***yè yé** nyé àmè sì nyá nú lè Xéxé lá mè.
in Okon POSS mind inside TOP **3SG FOC** / **LOGP FOC** is person REL know thing LOC world DEF inside
'In Okon's mind, he is the smartest person in the world.'

Constructions akin to (110a) produce similar results in Yoruba (111) and Igbo (112). While the ordinary pronoun is licensed, the logophor was not fully accepted by our speakers.

(111) *B&I context (108) in Yoruba*

a. Ni ọkan Kọ́la, **ó** nigbagbo wipe òun gbon.
in mind Kola **3SG** believe that LOG smart
'In Kola's mind, he believes he is smart.'

b.??Ni ọkan Kọ́la, **òun** nigbagbo wipe òun gbon.
in mind Kola **LOG** believe that LOG smart
'In Kola's mind, he believes he is smart.'
Comment: It is marked. In fact, speakers might have difficulty parsing it and connecting the reference.

(112) *B&I context (108) in Igbo*

a. N'uche Okon, **ọ́** kwere ne yá ma ihe.
PREP=mind Okon **3SG** believe that LOG know thing
'In Okon's mind, he believes he is smart.'

b.??N'uche Okon, **yá** kwere ne yá ma ihe.
PREP=mind Okon **LOG** believe that LOG know thing
'In Kola's mind, he believes he is smart.'
Comment: It sounds degraded to me.

[45] The unacceptability of *yè* in (110b) is not due to the fact that the logophor is accompanied by a focus marker since a focused logophor is in principle possible, see the example in (109), taken from Bimpeh (2023: 162).

(109) Mary be **yè yé** dzó.
Mary say **LOGP FOC** leave
'Mary said it was she who left.'

In light of such data, a proposal along the lines of Charnavel (2020a,b) would have to explain why the silent logophoric operator, proposed in (102), which (locally) binds English/French/Japanese exempt anaphora cannot appear in the same environments to bind the LOGP in Ewe/Yoruba/Igbo. We leave the development of a unifying theory to future work.